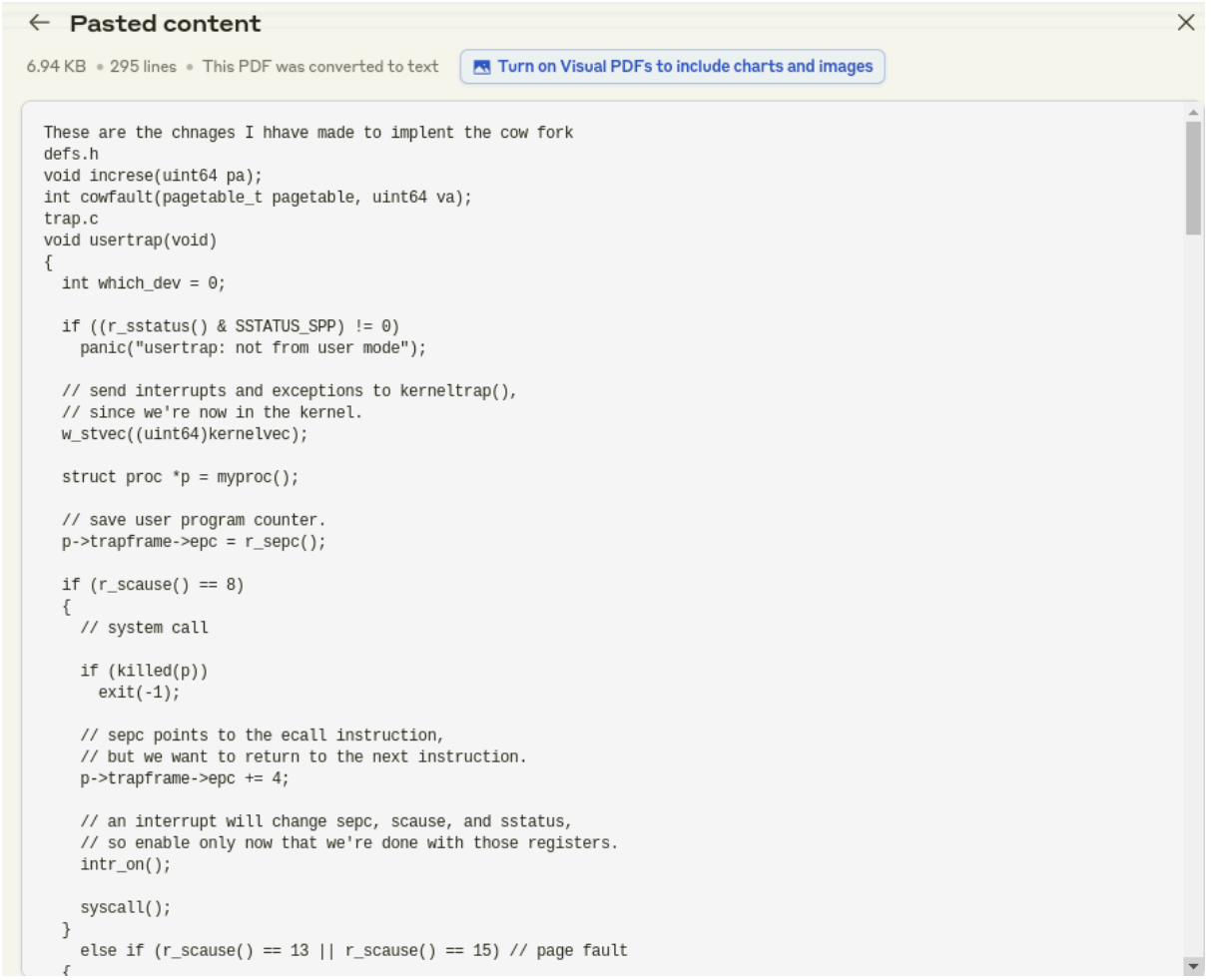
AI prompts and responses (MP- 3)

1. LAZY Fork

   Claude



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```
These are the chnages I hhave made to implent the cow fork
defs.h
void increse(uint64 pa);
int cowfault(pagetable_t pagetable, uint64 va);
trap.c
void usertrap(void)
{
  int which_dev = 0;

  if ((r_sstatus() & SSTATUS_SPP) != 0)
    panic("usertrap: not from user mode");

  // send interrupts and exceptions to kerneltrap(),
  // since we're now in the kernel.
  w_stvec((uint64)kernelvec);

  struct proc *p = myproc();

  // save user program counter.
  p->trapframe->epc = r_sepc();

  if (r_scause() == 8)
  {
    // system call

    if (killed(p))
      exit(-1);

    // sepc points to the ecall instruction,
    // but we want to return to the next instruction.
    p->trapframe->epc += 4;

    // an interrupt will change sepc, scause, and sstatus,
    // so enable only now that we're done with those registers.
    intr_on();

    syscall();
  }
    else if (r_scause() == 13 || r_scause() == 15) // page fault
  {
```

```
    syscall();
  }
    else if (r_scause() == 13 || r_scause() == 15) // page fault
  {
    // Handle Copy-On-Write page fault
    if (cowfault(p->pagetable, r_stval()) < 0)
    {
      // If cowfault fails, kill the process
      setkilled(p);
    }
  }
  else if ((which_dev = devintr()) != 0)
  {
    // ok
  }
  else
  {
    printf("usertrap(): unexpected scause %p pid=%d\n", r_scause(), p->pid);
    printf("            sepc=%p stval=%p\n", r_sepc(), r_stval());
    setkilled(p);
  }

  if (killed(p))
    exit(-1);

  // give up the CPU if this is a timer interrupt.
  if (which_dev == 2)
    yield();

  usertrapret();
}
kalloc.c
// Physical memory allocator, for user processes,
// kernel stacks, page-table pages,
// and pipe buffers. Allocates whole 4096-byte pages.

#include "types.h"
#include "param.h"
#include "memlayout.h"
#include "spinlock.h"
#include "riscv.h"
```


```
    syscall();
  }
    else if (r_scause() == 13 || r_scause() == 15) // page fault
  {
    // Handle Copy-On-Write page fault
    if (cowfault(p->pagetable, r_stval()) < 0)
    {
      // If cowfault fails, kill the process
      setkilled(p);
    }
  }
  else if ((which_dev = devintr()) != 0)
  {
    // ok
  }
  else
  {
    printf("usertrap(): unexpected scause %p pid=%d\n", r_scause(), p->pid);
    printf("            sepc=%p stval=%p\n", r_sepc(), r_stval());
    setkilled(p);
  }

  if (killed(p))
    exit(-1);

  // give up the CPU if this is a timer interrupt.
  if (which_dev == 2)
    yield();

  usertrapret();
}
kalloc.c
// Physical memory allocator, for user processes,
// kernel stacks, page-table pages,
// and pipe buffers. Allocates whole 4096-byte pages.

#include "types.h"
#include "param.h"
#include "memlayout.h"
#include "spinlock.h"
#include "riscv.h"
```

```
void freerange(void *pa_start, void *pa_end);

extern char end[]; // first address after kernel.
                   // defined by kernel.ld.

struct run {
  struct run *next;
};

struct {
  struct spinlock lock;
  struct run *freelist;
} kmem;

// * Define the reference count array at the top of the file *
int refcnt[PHYSTOP / PGSIZE];

void
kinit()
{
  initlock(&kmem.lock, "kmem");
  freerange(end, (void*)PHYSTOP);
}

void freerange(void *pa_start, void *pa_end)
{
  char *p;
  p = (char *)PGROUNDUP((uint64)pa_start);
  for (; p + PGSIZE <= (char *)pa_end; p += PGSIZE)
  {
    refcnt[(uint64)p / PGSIZE] = 1;
    kfree(p);
  }
}

// Free the page of physical memory pointed at by pa,
// which normally should have been returned by a
// call to kalloc().  (The exception is when
// initializing the allocator; see kinit above.)
void kfree(void *pa)
```

```
// call to kalloc().  (The exception is when
// initializing the allocator; see kinit above.)
void kfree(void *pa)
{
  struct run *r;
  r = (struct run *)pa;
  if (((uint64)pa % PGSIZE) != 0 || (char *)pa < end || (uint64)pa >= PHYSTOP)
    panic("kfree");
    //when we free the page decraese the refcnt of the pa
    //we need to acquire the lock
    //and get the really current cnt for the current fucntion
  acquire(&kmem.lock);
  int pn = (uint64)r / PGSIZE;
  if (refcnt[pn] < 1)
    panic("kfree panic");
  refcnt[pn] -= 1;
  int tmp = refcnt[pn];
  release(&kmem.lock);

  if (tmp >0)
    return;
  // Fill with junk to catch dangling refs.
  memset(pa, 1, PGSIZE);

  acquire(&kmem.lock);
  r->next = kmem.freelist;
  kmem.freelist = r;
  release(&kmem.lock);
}
// Allocate one 4096-byte page of physical memory.
// Returns a pointer that the kernel can use.
// Returns 0 if the memory cannot be allocated.
void *
kalloc(void)
{
  struct run *r;

  acquire(&kmem.lock);
  r = kmem.freelist;

  if (r)
```

```
  r = kmem.freelist;

  if (r)
  {
    int pn = (uint64)r / PGSIZE;
    if(refcnt[pn]!=0){
      panic("refcnt kalloc");
    }
    refcnt[pn] = 1;
    kmem.freelist = r->next;
  }

  release(&kmem.lock);

  if (r)
    memset((char *)r, 5, PGSIZE); // fill with junk
  return (void *)r;
}


void increse(uint64 pa)
{
    //acquire the lock
  acquire(&kmem.lock);
  int pn = pa / PGSIZE;
  if(pa>PHYSTOP || refcnt[pn]<1){
    panic("increase ref cnt");
  }
  refcnt[pn]++;
  release(&kmem.lock);
}
vm.c
/ frees any allocated pages on failure.
int uvmcopy(pagetable_t old, pagetable_t new, uint64 sz)
{
  pte_t *pte;
  uint64 pa, i;
  uint flags;

  for (i = 0; i < sz; i += PGSIZE)
```

```
  {
    if ((pte = walk(old, i, 0)) == 0)
      panic("uvmcopy: pte should exist");
    if ((*pte & PTE_V) == 0)
      panic("uvmcopy: page not present");
    // fix the permission bits
    pa = PTE2PA(*pte);
    *pte &= ~PTE_W;
    flags = PTE_FLAGS(*pte);
    // not allocated
    //  if((mem = kalloc()) == 0)
    //    goto err;
    //  memmove(mem, (char*)pa, PGSIZE);
    // increase refcnt
    increse(pa);
    // map the va to the same pa using flags
    if (mappages(new, i, PGSIZE, (uint64)pa, flags) != 0)
    {
      goto err;
    }
  }
  return 0;

err:
  uvmunmap(new, 0, i / PGSIZE, 1);
  return -1;
}
int cowfault(pagetable_t pagetable, uint64 va)
{
  if (va >= MAXVA)
    return -1;
  pte_t *pte = walk(pagetable, va, 0);
  if (pte == 0)
    return -1;
  if ((*pte & PTE_U) == 0 || (*pte & PTE_V) == 0)
    return -1;
  uint64 pa1 = PTE2PA(*pte);
  uint64 pa2 = (uint64)kalloc();
  if (pa2 == 0)
  {
```

```
  if (pa2 == 0)
  {
    // panic("cow panic kalloc");
    return -1;
  }

  memmove((void *)pa2, (void *)pa1, PGSIZE);
  *pte = PA2PTE(pa2) | PTE_U | PTE_V | PTE_W | PTE_X | PTE_R;
  kfree((void *)pa1);
  return 0;
}
// Copy from kernel to user.
// Copy len bytes from src to virtual address dstva in a given page table.
// Return 0 on success, -1 on error.
int copyout(pagetable_t pagetable, uint64 dstva, char *src, uint64 len)
{
  uint64 n, va0, pa0;

  while (len > 0)
  {
    va0 = PGROUNDDOWN(dstva);
    pa0 = walkaddr(pagetable, va0);
    if (pa0 == 0)
      return -1;

    // Check if the page is writable. If not, handle it as a COW page.
    pte_t *pte = walk(pagetable, va0, 0);
    if (pte == 0)
      return -1;

    if (!(*pte & PTE_W))
    {
      // Call cowfault to handle the copy-on-write
      if (cowfault(pagetable, va0) < 0)
        return -1;

      // After cowfault, retrieve the updated physical address
      pa0 = walkaddr(pagetable, va0);
      if (pa0 == 0)
        return -1;
    }
```

```
    if (pte == 0)
      return -1;

    if (!(*pte & PTE_W))
    {
      // Call cowfault to handle the copy-on-write
      if (cowfault(pagetable, va0) < 0)
        return -1;

      // After cowfault, retrieve the updated physical address
      pa0 = walkaddr(pagetable, va0);
      if (pa0 == 0)
        return -1;
    }

    n = PGSIZE - (dstva - va0);
    if (n > len)
      n = len;

    memmove((void *)(pa0 + (dstva - va0)), src, n);

    len -= n;
    src += n;
    dstva = va0 + PGSIZE;
  }
  return 0;
}
Based on this tell me how I can do
Part 2: Copy-On-Write (COW) Fork Performance Analysis
Page Fault Frequency:
Record the frequency of page faults during the operation of the COW fork. Test it with processes that read only, as well a
s those that modify memory.
The boss is particularly interested in knowing how many times the COW mechanism is triggered under different scenarios.
Brief Analysis:
Discuss the benefits of COW fork in terms of efficiency and memory conservation and provide a few sentences on any areas w
here COW could be further optimized.
with this hands-down report, we'll give the boss the essential insights they're looking for without going overboard. Once
it's done, we can get back to our usual lazy routine. So, roll up your sleeves—this time, no shortcuts! It's all about wor
king smart and delivering concise, impactful results!
```

COW Fork Performance Test Program

```c
#include "kernel/types.h"
#include "kernel/stat.h"
#include "user/user.h"

#define PAGES 10
#define PAGE_SIZE 4096
#define TEST_ITERATIONS 5

// Global counter for page faults - will need to be updated in trap.c
int page_fault_count = 0;

void test_read_only() {
  printf("Read-only test starting...\n");

  // Allocate memory
  char *memory = malloc(PAGES * PAGE_SIZE);
  if(memory == 0) {
    printf("Failed to allocate memory\n");
    exit(1);
  }

  // Initialize memory with some data
  for(int i = 0; i < PAGES * PAGE_SIZE; i++) {
    memory[i] = 'A';
  }

  // Reset page fault counter before fork
  page_fault_count = 0;
```

COW Fork Performance Test Program

```
  // Reset page fault counter before fork
  page_fault_count = 0;

  int pid = fork();
  if(pid < 0) {
    printf("Fork failed\n");
    exit(1);
  }

  if(pid == 0) {
    // Child process - only read memory
    char sum = 0;
    for(int i = 0; i < PAGES * PAGE_SIZE; i++) {
      sum += memory[i];
    }
    printf("Child read-only test complete. Sum: %d\n", sum);
    exit(0);
  } else {
    wait(0);
    printf("Read-only test page faults: %d\n", page_fault_count);
  }

  free(memory);
}

void test_write_heavy() {
  printf("\nWrite-heavy test starting...\n");

  char *memory = malloc(PAGES * PAGE_SIZE);
```

```
  free(memory);
}

void test_write_heavy() {
  printf("\nWrite-heavy test starting...\n");

  char *memory = malloc(PAGES * PAGE_SIZE);
  if(memory == 0) {
    printf("Failed to allocate memory\n");
    exit(1);
  }

  // Initialize memory
  for(int i = 0; i < PAGES * PAGE_SIZE; i++) {
    memory[i] = 'B';
  }

  // Reset page fault counter
  page_fault_count = 0;

  int pid = fork();
  if(pid < 0) {
    printf("Fork failed\n");
    exit(1);
  }

  if(pid == 0) {
    // Child process - write to every page
    for(int i = 0; i < PAGES; i++) {
      memory[i * PAGE_SIZE] = 'C';  // Write to first byte of each page
    }
    printf("Child write test complete\n");
    exit(0);
  } else {
    wait(0);
    printf("Write-heavy test page faults: %d\n", page_fault_count);
  }

  free(memory);
}

int main() {
  printf("Starting COW Fork Performance Tests\n");
  printf("Testing with %d pages of %d bytes each\n\n", PAGES, PAGE_SIZE);

  // Run multiple iterations of each test
  for(int i = 0; i < TEST_ITERATIONS; i++) {
    printf("Iteration %d:\n", i+1);
    test_read_only();
    test_write_heavy();
    printf("\n");
  }

  return 0;
}
```

To implement this test and get the measurements:

1. First, add a global counter in trap.c:

c Copy

```c
// At the top of trap.c
int page_fault_count = 0;

// In the page fault handling section of usertrap(), increment the counter:
else if (r_scause() == 13 || r_scause() == 15) {
  page_fault_count++;  // Add this line
  if (cowfault(p->pagetable, r_stval()) < 0) {
    setkilled(p);
  }
}
```

2. Add the counter to user.h:

c Copy

```c
extern int page_fault_count;
```

3. Compile and run the test program. Here's a sample report format for your boss:

P where hsould thhe test pprogram be preset? in kernel or user? what should it be named? how hhsould use it for testing? is thhere any other chnages I should make ij the xv6 code?

Let me clarify the test program setup and additional changes needed.

1. First, let's place the test program in the user directory:

</> **User Test Program (user/cowtest.c)**
Click to open code

2. Add page fault tracking in kernel/trap.c:

</> **Kernel Trap Changes (kernel/trap.c)**
Click to open code

3. Modify Makefile to include the test program:

makefile Copy

```
# Add cowtest to UPROGS in Makefile
UPROGS=\
  $U/_cat\
  $U/_echo\
  $U/_forktest\
  $U/_grep\
```

```
$U/_grind\
$U/_wc\
$U/_zombie\
$U/_cowtest\    # Add this line
```

To run the tests:

1. Build xv6 with the new changes:

```bash
make clean
make qemu
```

2. In the xv6 shell, run the test:

```bash
cowtest
```

The test will output:

- Number of page faults for read-only operations
- Number of page faults for write operations
- Time taken for each test (in ticks)

Additional suggestions:

1. You might want to add a counter in kalloc.c to track memory allocations:

1. You might want to add a counter in kalloc.c to track memory allocations:

c Copy

```c
// Add in kalloc.c
uint64 total_allocations = 0;

// Modify kalloc():
void *
kalloc(void)
{
  struct run *r;

  acquire(&kmem.lock);
  r = kmem.freelist;
  if(r) {
    total_allocations++;  // Track allocations
    kmem.freelist = r->next;
  }
  release(&kmem.lock);

  if(r)
    memset((char*)r, 5, PGSIZE); // fill with junk
  return (void*)r;
}
```

The test program will help you measure:

1. How often COW page faults occur

The test program will help you measure:

1. How often COW page faults occur
2. Memory usage efficiency
3. Performance difference between read-only and write-heavy workloads

When running tests:

1. Start with small number of pages (10) to verify everything works
2. Then try larger numbers (100, 1000) to stress test
3. Compare performance with and without writes
4. Look for patterns in page fault frequency

The output will help you analyze:

- How well COW works for read-heavy workloads
- The overhead of page faults in write-heavy scenarios
- Memory efficiency gains from sharing pages

Would you like me to explain any part in more detail or help with additional test cases?

ⓘ Claude does not have the ability to run the code it generates yet.

P ls user alarmtest.c init.c ls.sym stressfs.sym *cat initcode *mkdir ulib.c cat.asm initcode.asm mkdir.asm ulib.d cat.c initcode.d mkdir.c ulib.o cat.d initcode.o mkdir.d umalloc.c cat.o initcode.out mkdir.o umalloc.d cat.sym initcode.S mkdir.sym umalloc.o cowtest.c UPROGS=\ $U/_cat\ $U/_echo\ $U/_forktest\ $U/_grep\ $U/_init\ $U/_kill\ $U/_ln\ $U/_ls\ $U/_mkdir\

```
  int pid = fork();
  if(pid < 0) {
    printf("Fork failed\n");
    exit(1);
  }

  if(pid == 0) {
    // Child process - only read memory
    char sum = 0;
    for(int i = 0; i < PAGES * PAGE_SIZE; i++) {
      sum += memory[i];
    }
    printf("Child read-only test complete. Sum: %d\n", sum);
    exit(0);
  } else {
    wait(0);
    int end_ticks = uptime();
    printf("Read-only test completed in %d ticks\n", end_ticks - start_ticks);
  }

  free(memory);
}

void test_write_heavy() {
  printf("\nWrite-heavy test starting...\n");

  char *memory = malloc(PAGES * PAGE_SIZE);
  if(memory == 0) {
```

```
  char *memory = malloc(PAGES * PAGE_SIZE);
  if(memory == 0) {
    printf("Failed to allocate memory\n");
    exit(1);
  }

  // Initialize memory
  for(int i = 0; i < PAGES * PAGE_SIZE; i++) {
    memory[i] = 'B';
  }

  int start_ticks = uptime();  // Get starting time

  int pid = fork();
  if(pid < 0) {
    printf("Fork failed\n");
    exit(1);
  }

  if(pid == 0) {
    // Child process - write to every page
    for(int i = 0; i < PAGES; i++) {
      memory[i * PAGE_SIZE] = 'C';  // Write to first byte of each page
    }
    printf("Child write test complete\n");
    exit(0);
  } else {
    wait(0);
```

```
    for(int i = 0; i < PAGES; i++) {
      memory[i * PAGE_SIZE] = 'C';  // Write to first byte of each page
    }
    printf("Child write test complete\n");
    exit(0);
  } else {
    wait(0);
    int end_ticks = uptime();
    printf("Write-heavy test completed in %d ticks\n", end_ticks - start_ticks);
  }

  free(memory);
}

int main() {
  printf("Starting COW Fork Performance Tests\n");
  printf("Testing with %d pages of %d bytes each\n\n", PAGES, PAGE_SIZE);

  // Run multiple iterations of each test
  for(int i = 0; i < TEST_ITERATIONS; i++) {
    printf("Iteration %d:\n", i+1);
    test_read_only();
    test_write_heavy();
    printf("\n");
  }

  return 0;
}
```

← Kernel Trap Changes (kernel/trap.c)

```
// Add at the top of trap.c with other globals
uint page_faults = 0;  // Track total page faults

// Modify the page fault handling section in usertrap():
else if (r_scause() == 13 || r_scause() == 15) {
  page_faults++;  // Increment counter
  printf("pid %d: page fault %d at addr %p\n",
         p->pid, page_faults, r_stval());
  if (cowfault(p->pagetable, r_stval()) < 0) {
    setkilled(p);
  }
}
```

P

I am trying to the following in xv6

* Record the **frequency of page faults** during the operation of the COW fork. Test it with processes that read only, as well as those that modify memory.

I have modified vm.c

```
* int cowfault(pagetable_t pagetable, uint64 va)
{
 if (va >= MAXVA)
  return -1;
 pte_t pte = walk(pagetable, va, 0);
 if (pte == 0)
  return -1;
 if ((pte & PTE_U) == 0 || (pte & PTE_V) == 0)
  return -1;
 uint64 pa1 = PTE2PA(pte);
 uint64 pa2 = (uint64)kalloc();
 if (pa2 == 0)
 {
  // panic("cow panic kalloc");
  return -1;
 }

 memmove((void *)pa2, (void *)pa1, PGSIZE);
 *pte = PA2PTE(pa2) | PTE_U | PTE_V | PTE_W | PTE_X | PTE_R;
 kfree((void *)pa1);

 // Increment the COW fault counter safely
 acquire(&kmem.lock);    // Acquire lock to prevent race conditions
 cow_fault_count++;      // Increment COW fault count
```

```
}
* // Return 0 on success, -1 on error.
int copyout(pagetable_t pagetable, uint64 dstva, char *src, uint64 len)
{
 uint64 n, va0, pa0;

 while (len > 0)
 {
  va0 = PGROUNDDOWN(dstva);
  pa0 = walkaddr(pagetable, va0);
  if (pa0 == 0)
   return -1;

  // Check if the page is writable. If not, handle it as a COW page.
  pte_t *pte = walk(pagetable, va0, 0);
  if (pte == 0)
   return -1;

  if (!(*pte & PTE_W))
  {
   // Call cowfault to handle the copy-on-write
   if (cowfault(pagetable, va0) < 0)
    return -1;

   // After cowfault, retrieve the updated physical address
   pa0 = walkaddr(pagetable, va0);
   if (pa0 == 0)
    return -1;
  }

  n = PGSIZE - (dstva - va0);
```

```
      return -1;
    // After cowfault, retrieve the updated physical address
    pa0 = walkaddr(pagetable, va0);
    if (pa0 == 0)
     return -1;
   }

   n = PGSIZE - (dstva - va0);
   if (n > len)
    n = len;

   memmove((void *)(pa0 + (dstva - va0)), src, n);

   len -= n;
   src += n;
   dstva = va0 + PGSIZE;
  }
  return 0;
}
* trap.c
* void usertrap(void)
{
  int which_dev = 0;

  if ((r_sstatus() & SSTATUS_SPP) != 0)
   panic("usertrap: not from user mode");

  // send interrupts and exceptions to kerneltrap(),
  // since we're now in the kernel.
  w_stvec((uint64)kernelvec);

  struct proc *p = myproc();
```

```
    release(&kmem.lock);
  // Handle Copy-On-Write page fault
  if (cowfault(p->pagetable, r_stval()) < 0)
  {
    // If cowfault fails, kill the process
    setkilled(p);
  }
 }
 else if ((which_dev = devintr()) != 0)
 {
  // ok
 }
 else
 {
  printf("usertrap(): unexpected scause %p pid=%d\n", r_scause(), p->pid);
  printf("            sepc=%p stval=%p\n", r_sepc(), r_stval());
  setkilled(p);
 }

 if (killed(p))
  exit(-1);

 // give up the CPU if this is a timer interrupt.
 if (which_dev == 2)
  yield();

 usertrapret();
}
* kalloc.c
*
```

```
}
* kalloc.c
*
// Free the page of physical memory pointed at by pa,
// which normally should have been returned by a
// call to kalloc().  (The exception is when
// initializing the allocator; see kinit above.)
void kfree(void *pa)
{
 struct run *r;
 r = (struct run *)pa;
 if (((uint64)pa % PGSIZE) != 0 || (char *)pa < end || (uint64)pa >= PHYSTOP)
  panic("kfree");
  //when we free the page decraese the refcnt of the pa
  //we need to acquire the lock
  //and get the really current cnt for the current fucntion
 acquire(&kmem.lock);
 int pn = (uint64)r / PGSIZE;
 if (refcnt[pn] < 1)
  panic("kfree panic");
 refcnt[pn] -= 1;
 int tmp = refcnt[pn];
 release(&kmem.lock);

 if (tmp >0)
  return;
 // Fill with junk to catch dangling refs.
 memset(pa, 1, PGSIZE);
```

```
  return;
 // Fill with junk to catch dangling refs.
 memset(pa, 1, PGSIZE);

 acquire(&kmem.lock);
 r->next = kmem.freelist;
 kmem.freelist = r;
 release(&kmem.lock);
}
// Allocate one 4096-byte page of physical memory.
// Returns a pointer that the kernel can use.
// Returns 0 if the memory cannot be allocated.
void *
kalloc(void)
{
 struct run *r;

 acquire(&kmem.lock);
 r = kmem.freelist;

 if (r)
 {
  int pn = (uint64)r / PGSIZE;
  if(refcnt[pn]!=0){
   panic("refcnt kalloc");
  }
  refcnt[pn] = 1;
  kmem.freelist = r->next;
 }

 release(&kmem.lock);
```

1. Maintain a page fault counter:

   - In `trap.c`, in the `usertrap()` function, you're already incrementing the `total_fault_count` for every page fault. However, you'll need to specifically track the COW-related page faults.
   - Create a new global variable `cow_fault_count` in `defs.h` to keep track of the COW-related page faults.
   - In the `usertrap()` function, after the call to `cowfault()`, if it succeeds, increment the `cow_fault_count` variable. This will give you the count of COW-related page faults.

2. **Protect the `cow_fault_count` variable:**

   - To ensure thread-safety and avoid race conditions when incrementing the `cow_fault_count`, you'll need to acquire and release the `kmem.lock` when updating the variable.
   - In the `usertrap()` function, wrap the increment of `cow_fault_count` with a lock acquisition and release, similar to how you're protecting the `total_fault_count`.

3. **Test with read-only and modifying processes:**

   - Create two test processes, one that only reads memory and another that modifies memory.
   - For the read-only process, you can use the existing `user_tests` in `usertests.c`. These tests should trigger COW page faults, but the `cow_fault_count` should remain low compared to the modifying process.

- For the read-only process, you can use the existing `user_tests` in `usertests.c`. These tests should trigger COW page faults, but the `cow_fault_count` should remain low compared to the modifying process.
- For the modifying process, you can create a new test that repeatedly writes to different memory locations, which should trigger more COW page faults and increase the `cow_fault_count`.

4. **Display the page fault counts**:
   - In your test code, add logic to print the `total_fault_count` and `cow_fault_count` values after the test processes have completed.
   - This will allow you to verify that the page fault counting is working as expected.

Here's an example of how you can update the `usertrap()` function in `trap.c` to track the COW-related page faults:

c Copy

```c
void usertrap(void)
{
  // ... (existing code)

  if (r_scause() == 13 || r_scause() == 15) // page fault
  {
    // Increment total page fault counter
    acquire(&kmem.lock);
    total_fault_count++;      // Increment total fault count
    release(&kmem.lock);
```

```
    acquire(&kmem.lock);
    total_fault_count++;      // Increment total fault count
    release(&kmem.lock);

    // Handle Copy-On-Write page fault
    if (cowfault(p->pagetable, r_stval()) == 0)
    {
      // Increment COW fault counter safely
      acquire(&kmem.lock);
      cow_fault_count++;      // Increment COW fault count
      release(&kmem.lock);
    }
    else
    {
      // If cowfault fails, kill the process
      setkilled(p);
    }
  }

  // ... (existing code)
}
```

Copy

Make sure to declare the `cow_fault_count` variable in `defs.h` and initialize it to 0. You'll also need to add the appropriate print statements in your test code to display the `total_fault_count` and `cow_fault_count` values.

With these changes, you should be able to track the page fault frequencies for read-only and modifying processes and observe the difference in the COW-related page faults.

P is this correct?

```
void usertrap(void)
{
  int which_dev = 0;

  if ((r_sstatus() & SSTATUS_SPP) != 0)
    panic("usertrap: not from user mode");

  // send interrupts and exceptions to kerneltrap(),
  // since we're now in the kernel.
  w_stvec((uint64)kernelvec);

  struct proc *p = myproc();

  // save user program counter.
  p->trapframe->epc = r_sepc();

  if (r_scause() == 8)
  {
    // system call

    if (killed(p))
      exit(-1);

    // sepc points to the ecall instruction,
    // but we want to return to the next instruction.
    p->trapframe->epc += 4;

    // an interrupt will change sepc, scause, and sstatus,
    // so enable only now that we're done with those registers.
    intr_on();

    syscall();
```

```
  // so enable only now that we're done with those registers.
  intr_on();

  syscall();
 }
 if (r_scause() == 13 || r_scause() == 15) // page fault
 {
  // Increment total page fault counter
  acquire(&kmem.lock);
  total_fault_count++; // Increment total fault count
  release(&kmem.lock);

  // Handle Copy-On-Write page fault
  if (cowfault(p->pagetable, r_stval()) == 0)
  {
   // Increment COW fault counter safely
   acquire(&kmem.lock);
   cow_fault_count++; // Increment COW fault count
   release(&kmem.lock);
  }
  else if(cowfault(p->pagetable, r_stval()) < 0)
  {
   // If cowfault fails, kill the process
   setkilled(p);
  }
 }
 else if ((which_dev = devintr()) != 0)
 {
  // ok
```

~~kernel/trap.c:80:7: error: 'cow_fault_count' undeclared (first use in this function)~~

~~80 | cow_fault_count++; // Increment COW fault count~~

| ^~~~~~~~~~~~~~~

make: *** [<builtin>: kernel/trap.o] Error 1

The code you provided looks correct in terms of the logic for incrementing the `total_fault_count` and `cow_fault_count` in the `usertrap()` function. However, the errors you're encountering are related to the variables being undeclared.

To fix this, you'll need to make the following changes:

1. **Declare the necessary variables:**
   - In `defs.h`, declare the variables `total_fault_count` and `cow_fault_count` as global variables:

```c
extern int total_fault_count;
extern int cow_fault_count;
```

2. **Initialize the variables:**
   - In `proc.c`, initialize the `total_fault_count` and `cow_fault_count` to 0 in the `initproc()` function:

```c
void
initproc(void)
{
```

readability and maintainability of your code:

1. **Separate the page fault handling logic:**
   - Create a new function, say `handle_page_fault()`, that encapsulates the logic for handling page faults, including the COW-related checks and updates.
   - This will make the `usertrap()` function more concise and easier to understand.
2. **Use consistent variable names:**
   - The variable name `which_dev` is a bit vague. Consider using a more descriptive name, such as `interrupt_type` or `device_number`.
3. **Add comments:**
   - Add comments to explain the purpose and functionality of the different sections of the `usertrap()` function. This will make the code more readable and easier to maintain.

By making these changes, your code will be more organized, easier to understand, and less prone to errors.

ⓘ Claude does not have the ability to run the code it generates yet.

P

```
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -MD
-mcmodel=medany -ffreestanding -fno-common -nostdlib -mno-relax -I. -fno-stack-
protector -fno-pie -no-pie   -c -o kernel/kalloc.o kernel/kalloc.c
kernel/kalloc.c:24:3: error: conflicting types for 'kmem'; have 'struct <anonymous>'
  24 | } kmem;
```

```
#include "defs.h"
// handle an interrupt, exception, or system call from user space.
// called from trampoline.S
//
void usertrap(void)
{
  int which_dev = 0;

  if ((r_sstatus() & SSTATUS_SPP) != 0)
    panic("usertrap: not from user mode");

  // send interrupts and exceptions to kerneltrap(),
  // since we're now in the kernel.
  w_stvec((uint64)kernelvec);

  struct proc *p = myproc();

  // save user program counter.
  p->trapframe->epc = r_sepc();

  if (r_scause() == 8)
  {
    // system call

    if (killed(p))
      exit(-1);

    // sepc points to the ecall instruction,
    // but we want to return to the next instruction.
    p->trapframe->epc += 4;

    // an interrupt will change sepc, scause, and sstatus,
    // so enable only now that we're done with those registers.
```

```
}
if (r_scause() == 13 || r_scause() == 15) // page fault
{
  // Increment total page fault counter
  acquire(&kmem.lock);
  total_fault_count++; // Increment total fault count
  release(&kmem.lock);

  // Handle Copy-On-Write page fault
  if (cowfault(p->pagetable, r_stval()) == 0)
  {
    // Increment COW fault counter safely
    acquire(&kmem.lock);
    cow_fault_count++; // Increment COW fault count
    release(&kmem.lock);
  }
  else
  {
    // If cowfault fails, kill the process
    setkilled(p);
  }
}
else if ((which_dev = devintr()) != 0)
{
  // ok
}
else
{
```

```
  else
  {
    printf("usertrap(): unexpected scause %p pid=%d\n", r_scause(), p->pid);
    printf("            sepc=%p stval=%p\n", r_sepc(), r_stval());
    setkilled(p);
  }

  if (killed(p))
    exit(-1);

  // give up the CPU if this is a timer interrupt.
  if (which_dev == 2)
    yield();

  usertrapret();
}
```

defs.h

```
#include "spinlock.h" // make sure spinlock.h is included for struct spinlock definition

// Declare kmem as an extern struct
extern struct {
  struct spinlock lock;
  struct run *freelist;
} kmem;

extern int total_fault_count;
extern int cow_fault_count;

#endif // DEFS_H
```

kalloc.c

```
#include "defs.h"
```

```
#endif // DEFS_H
```

kalloc.c

```
#include "defs.h"
void *
kalloc(void)
{
  struct run *r;

  acquire(&kmem.lock);
  r = kmem.freelist;

  if (r)
  {
    int pn = (uint64)r / PGSIZE;
    if(refcnt[pn]!=0){
      panic("refcnt kalloc");
    }
    refcnt[pn] = 1;
    kmem.freelist = r->next;
  }

  release(&kmem.lock);

  if (r)
    memset((char *)r, 5, PGSIZE); // fill with junk
  return (void *)r;
}

void increse(uint64 pa)
{
    //acquire the lock
```

```
{
  //acquire the lock
 acquire(&kmem.lock);
 int pn = pa / PGSIZE;
 if(pa>PHYSTOP || refcnt[pn]<1){
  panic("increase ref cnt");
 }
 refcnt[pn]++;
 release(&kmem.lock);
}
```

The error you're encountering is due to a conflict in the declaration of the `kmem` structure. The issue is that the `kmem` structure is being declared both in `defs.h` and in `kalloc.c`.

To fix this, you should move the declaration of the `kmem` structure to a common location, such as `defs.h`, and then include `defs.h` in both `trap.c` and `kalloc.c`.

Here's the updated code:

1. In `defs.h`:

c Copy

```c
struct {
  struct spinlock lock;
  struct run *freelist;
} kmem;
```

```
} kmem;

extern int total_fault_count;
extern int cow_fault_count;
```

2. In `trap.c`:

c

```c
#include "defs.h"

void usertrap(void)
{
  // ... (existing code)
}
```

3. In `kalloc.c`:

c

```c
#include "defs.h"

void *
kalloc(void)
{
  // ... (existing code)
}

void increse(uint64 pa)
```

```
}

void increse(uint64 pa)
{
  // ... (existing code)
}
```

Copy

By moving the declaration of the `kmem` structure to `defs.h` and including `defs.h` in both `trap.c` and `kalloc.c`, you should be able to resolve the conflicting types error.

Additionally, you may want to consider the other suggestions I made in my previous response, such as:

1. Separating the page fault handling logic into a dedicated function.
2. Using more descriptive variable names.
3. Adding comments to improve the readability and maintainability of the code.

These changes will help make your code more organized and easier to understand.

proc.c

15.75 KB • 773 lines • This PDF was converted to text Turn on Visual PDFs to include charts and images

```
// Map it high in memory, followed by an invalid
// guard page.
void proc_mapstacks(pagetable_t kpgtbl)
{
  struct proc *p;

  for (p = proc; p < &proc[NPROC]; p++)
  {
    char *pa = kalloc();
    if (pa == 0)
      panic("kalloc");
    uint64 va = KSTACK((int)(p - proc));
    kvmmap(kpgtbl, va, (uint64)pa, PGSIZE, PTE_R | PTE_W);
  }
}

// initialize the proc table.
void procinit(void)
{
  struct proc *p;

  initlock(&pid_lock, "nextpid");
  initlock(&wait_lock, "wait_lock");
  for (p = proc; p < &proc[NPROC]; p++)
  {
    initlock(&p->lock, "proc");
    p->state = UNUSED;
    p->kstack = KSTACK((int)(p - proc));
  }
}

// Must be called with interrupts disabled,
// to prevent race with process being moved
// to a different CPU.
int cpuid()
{
  int id = r_tp();
  return id;
}

// Return this CPU's cpu struct.
```

```
  return id;
}

// Return this CPU's cpu struct.
// Interrupts must be disabled.
struct cpu *
mycpu(void)
{
  int id = cpuid();
  struct cpu *c = &cpus[id];
  return c;
}

// Return the current struct proc *, or zero if none.
struct proc *
myproc(void)
{
  push_off();
  struct cpu *c = mycpu();
  struct proc *p = c->proc;
  pop_off();
  return p;
}

int allocpid()
{
  int pid;

  acquire(&pid_lock);
  pid = nextpid;
  nextpid = nextpid + 1;
  release(&pid_lock);

  return pid;
}

// Look in the process table for an UNUSED proc.
// If found, initialize state required to run in the kernel,
// and return with p->lock held.
// If there are no free procs, or a memory allocation fails, return 0.
static struct proc *
```

```
// and return with p->lock held.
// If there are no free procs, or a memory allocation fails, return 0.
static struct proc *
allocproc(void)
{
  struct proc *p;

  for (p = proc; p < &proc[NPROC]; p++)
  {
    acquire(&p->lock);
    if (p->state == UNUSED)
    {
      goto found;
    }
    else
    {
      release(&p->lock);
    }
  }
  return 0;

found:
  p->pid = allocpid();
  p->state = USED;

  // Allocate a trapframe page.
  if ((p->trapframe = (struct trapframe *)kalloc()) == 0)
  {
    freeproc(p);
    release(&p->lock);
    return 0;
  }

  // An empty user page table.
  p->pagetable = proc_pagetable(p);
  if (p->pagetable == 0)
  {
    freeproc(p);
    release(&p->lock);
    return 0;
  }
```

```
    return 0;
  }

  // Set up new context to start executing at forkret,
  // which returns to user space.
  memset(&p->context, 0, sizeof(p->context));
  p->context.ra = (uint64)forkret;
  p->context.sp = p->kstack + PGSIZE;
  p->rtime = 0;
  p->etime = 0;
  p->ctime = ticks;
  return p;
}

// free a proc structure and the data hanging from it,
// including user pages.
// p->lock must be held.
static void
freeproc(struct proc *p)
{
  if (p->trapframe)
    kfree((void *)p->trapframe);
  p->trapframe = 0;
  if (p->pagetable)
    proc_freepagetable(p->pagetable, p->sz);
  p->pagetable = 0;
  p->sz = 0;
  p->pid = 0;
  p->parent = 0;
  p->name[0] = 0;
  p->chan = 0;
  p->killed = 0;
  p->xstate = 0;
  p->state = UNUSED;
}

// Create a user page table for a given process, with no user memory,
// but with trampoline and trapframe pages.
pagetable_t
```

```
proc_pagetable(struct proc *p)
{
  pagetable_t pagetable;

  // An empty page table.
  pagetable = uvmcreate();
  if (pagetable == 0)
    return 0;

  // map the trampoline code (for system call return)
  // at the highest user virtual address.
  // only the supervisor uses it, on the way
  // to/from user space, so not PTE_U.
  if (mappages(pagetable, TRAMPOLINE, PGSIZE,
               (uint64)trampoline, PTE_R | PTE_X) < 0)
  {
    uvmfree(pagetable, 0);
    return 0;
  }

  // map the trapframe page just below the trampoline page, for
  // trampoline.S.
  if (mappages(pagetable, TRAPFRAME, PGSIZE,
               (uint64)(p->trapframe), PTE_R | PTE_W) < 0)
  {
    uvmunmap(pagetable, TRAMPOLINE, 1, 0);
    uvmfree(pagetable, 0);
    return 0;
  }

  return pagetable;
}

// Free a process's page table, and free the
// physical memory it refers to.
void proc_freepagetable(pagetable_t pagetable, uint64 sz)
{
  uvmunmap(pagetable, TRAMPOLINE, 1, 0);
```

```
void proc_freepagetable(pagetable_t pagetable, uint64 sz)
{
  uvmunmap(pagetable, TRAMPOLINE, 1, 0);
  uvmunmap(pagetable, TRAPFRAME, 1, 0);
  uvmfree(pagetable, sz);
}

// a user program that calls exec("/init")
// assembled from ../user/initcode.S
// od -t xC ../user/initcode
uchar initcode[] = {
    0x17, 0x05, 0x00, 0x00, 0x13, 0x05, 0x45, 0x02,
    0x97, 0x05, 0x00, 0x00, 0x93, 0x85, 0x35, 0x02,
    0x93, 0x08, 0x70, 0x00, 0x73, 0x00, 0x00, 0x00,
    0x93, 0x08, 0x20, 0x00, 0x73, 0x00, 0x00, 0x00,
    0xef, 0xf0, 0x9f, 0xff, 0x2f, 0x69, 0x6e, 0x69,
    0x74, 0x00, 0x00, 0x24, 0x00, 0x00, 0x00, 0x00,
    0x00, 0x00, 0x00, 0x00};

// Set up first user process.
void userinit(void)
{
  struct proc *p;

  p = allocproc();
  initproc = p;

  // allocate one user page and copy initcode's instructions
  // and data into it.
  uvmfirst(p->pagetable, initcode, sizeof(initcode));
  p->sz = PGSIZE;

  // prepare for the very first "return" from kernel to user.
  p->trapframe->epc = 0;     // user program counter
  p->trapframe->sp = PGSIZE; // user stack pointer

  safestrcpy(p->name, "initcode", sizeof(p->name));
  p->cwd = namei("/");

  p->state = RUNNABLE;
```

- In `trap.c`, in the `usertrap()` function, you're already incrementing the `total_fault_count` for every page fault. However, you'll need to specifically track the COW-related page faults.
- Create a new global variable `cow_fault_count` in `defs.h` to keep track of the COW-related page faults.
- In the `usertrap()` function, after the call to `cowfault()`, if it succeeds, increment the `cow_fault_count` variable. This will give you the count of COW-related page faults.

2. **Protect the `cow_fault_count` variable:**
   - To ensure thread-safety and avoid race conditions when incrementing the `cow_fault_count`, you'll need to acquire and release the `kmem.lock` when updating the variable.
   - In the `usertrap()` function, wrap the increment of `cow_fault_count` with a lock acquisition and release, similar to how you're protecting the `total_fault_count`.
3. **Test with read-only and modifying processes:**
   - Create two test processes, one that only reads memory and another that modifies memory.
   - For the read-only process, you can use the existing `user_tests` in `usertests.c`. These tests should trigger COW page faults, but the `cow_fault_count` should remain low compared to the modifying process.
   - For the modifying process, you can create a new test that repeatedly writes to

- For the modifying process, you can create a new test that repeatedly writes to different memory locations, which should trigger more COW page faults and increase the `cow_fault_count`.

4. **Display the page fault counts:**
   - In your test code, add logic to print the `total_fault_count` and `cow_fault_count` values after the test processes have completed.
   - This will allow you to verify that the page fault counting is working as expected.

Here's an example of how you can update the `usertrap()` function in `trap.c` to track the COW-related page faults:

c Copy

```c
void usertrap(void)
{
  // ... (existing code)

  if (r_scause() == 13 || r_scause() == 15) // page fault
  {
    // Increment total page fault counter
    acquire(&kmem.lock);
    total_fault_count++;      // Increment total fault count
    release(&kmem.lock);

    // Handle Copy-On-Write page fault
```

```
    // Handle Copy-On-Write page fault
    if (cowfault(p->pagetable, r_stval()) == 0)
    {
      // Increment COW fault counter safely
      acquire(&kmem.lock);
      cow_fault_count++;      // Increment COW fault count
      release(&kmem.lock);
    }
    else
    {
      // If cowfault fails, kill the process
      setkilled(p);
    }
  }

  // ... (existing code)
}
```

Make sure to declare the `cow_fault_count` variable in `defs.h` and initialize it to 0. You'll also need to add the appropriate print statements in your test code to display the `total_fault_count` and `cow_fault_count` values.

With these changes, you should be able to track the page fault frequencies for read-only and modifying processes and observe the difference in the COW-related page faults.

```
  p->cwd = namei("/");

  p->state = RUNNABLE;

  release(&p->lock);
}

// Grow or shrink user memory by n bytes.
// Return 0 on success, -1 on failure.
int growproc(int n)
{
  uint64 sz;
  struct proc *p = myproc();

  sz = p->sz;
  if (n > 0)
  {
    if ((sz = uvmalloc(p->pagetable, sz, sz + n, PTE_W)) == 0)
    {
      return -1;
    }
  }
  else if (n < 0)
  {
    sz = uvmdealloc(p->pagetable, sz, sz + n);
  }
  p->sz = sz;
  return 0;
}

// Create a new process, copying the parent.
// Sets up child kernel stack to return as if from fork() system call.
int fork(void)
{
  int i, pid;
  struct proc *np;
  struct proc *p = myproc();

  // Allocate process.
  if ((np = allocproc()) == 0)
  {
```

```
// Allocate process.
if ((np = allocproc()) == 0)
{
  return -1;
}

// Copy user memory from parent to child.
if (uvmcopy(p->pagetable, np->pagetable, p->sz) < 0)
{
  freeproc(np);
  release(&np->lock);
  return -1;
}
np->sz = p->sz;

// copy saved user registers.
*(np->trapframe) = *(p->trapframe);

// Cause fork to return 0 in the child.
np->trapframe->a0 = 0;

// increment reference counts on open file descriptors.
for (i = 0; i < NOFILE; i++)
  if (p->ofile[i])
    np->ofile[i] = filedup(p->ofile[i]);
np->cwd = idup(p->cwd);

safestrcpy(np->name, p->name, sizeof(p->name));

pid = np->pid;

release(&np->lock);

acquire(&wait_lock);
np->parent = p;
release(&wait_lock);

acquire(&np->lock);
np->state = RUNNABLE;
release(&np->lock);
```

```
// since we're now in the kernel.
w_stvec((uint64)kernelvec);

struct proc *p = myproc();

// save user program counter.
p->trapframe->epc = r_sepc();

if (r_scause() == 8)
{
 // system call

 if (killed(p))
  exit(-1);

 // sepc points to the ecall instruction,
 // but we want to return to the next instruction.
 p->trapframe->epc += 4;

 // an interrupt will change sepc, scause, and sstatus,
 // so enable only now that we're done with those registers.
 intr_on();

 syscall();
}
 else if (r_scause() == 13 || r_scause() == 15) // page fault
{
 // Increment total page fault counter
    acquire(&kmem.lock);
    total_fault_count++;     // Increment total fault count
    release(&kmem.lock);
 // Handle Copy-On-Write page fault
```

ChatGpt
https://chatgpt.com/share/673222d6-a7a0-8001-ab87-7cc25a40cbff

https://chatgpt.com/share/67323322-7070-800e-9f27-10375096efa9

https://chatgpt.com/share/673233be-8020-800e-87a2-b6e32a3ad14d

2. LAZY Read-Write
   https://chatgpt.com/share/67322603-479c-8002-8b6a-00033aff8326

   1.

P we need to implement lazy. **Here's how LAZY works:**

* Each operation (READ, WRITE, DELETE) takes some time to process.
* There's a limit on how many users can access the same file concurrently (for example, if 1 person is writing to the file while 2 others are reading, that is to be counted as 3 concurrent accesses).
* Users may get impatient and cancel their requests if LAZY doesn't get around to them in time.

Your task is to simulate this scenario, handle multiple user requests, and manage the concurrency and patience limits. Don't worry, though—you'll be using the same laid-back style you learned while implementing the COW fork.

Input Format

You'll be provided the following information to simulate LAZY:

Copy

```
r w d
n c T
u_1 f_1 o_1 t_1
u_2 f_2 o_2 t_2
…
u_m f_m o_m t_m
STOP
```

This represents:

* `r`, `w`, `d`: The time taken for **READ**, **WRITE**, and **DELETE** operations (in seconds).
* `n`: The number of files (named file 1, file 2, ..., file n).
* `c`: The maximum number of users that can access a single file concurrently.
* `T`: The maximum time a user will wait for their request before they cancel it.
* `u_i`, `f_i`, `o_i`, `t_i`: Information about each user's request:
  * `u_i`: The ID of the user.
  * `f_i`: The file number they want to access.
  * `o_i`: The operation (READ, WRITE, DELETE) they want to perform.
  * `t_i`: The time (in seconds) when they make the request.

The input ends with the word `STOP`.

Simulation Rules

LAZY's behavior is defined as follows:

1. **Processing requests**:
   * LAZY waits for 1 second after a request arrives before starting to process it.

← Fixed LAZY Implementation

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <pthread.h>
#include <unistd.h>
#include <time.h>

#define MAX_USERS 1500
#define RESET "\033[0m"
#define YELLOW "\033[1;33m"
#define PINK "\033[1;35m"
#define WHITE "\033[1;37m"
#define GREEN "\033[1;32m"
#define RED "\033[1;31m"

typedef struct req {
    int user_id;
    int file_id;
    char operation[10];
    int start_time;
    int req_time;
    int completion_time;
} req;

typedef struct file {
    int readers;
    int writers;
    int deleted;
    int pending_delete;  // New flag to track pending delete operations
    pthread_mutex_t lock;
    pthread_cond_t cond;
} file;

typedef struct ThreadArgs {
    req *request;
    file *file_states;
} ThreadArgs;

int r, w, d;
int n, c, t;
req requests[MAX_USERS];
int request_count = 0;
pthread_mutex_t print_mutex;
int current_time = 0;

// Print functions remain the same as in your previous code...
void print_request(int user_id, const char* operation, int file_id, int time) {
    pthread_mutex_lock(&print_mutex);
```

← Fixed LAZY Implementation

```
    pthread_mutex_unlock(&print_mutex);
}

void print_completed(int user_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(GREEN "The request for User %d was completed at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_canceled(int user_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(RED "User %d canceled the request due to no response at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

// Helper function to check if any read/write operations are waiting
int has_pending_read_write(file *file_state) {
    for (int i = 0; i < request_count; i++) {
        if (requests[i].file_id - 1 == (file_state - file_state) &&
            (strcmp(requests[i].operation, "READ") == 0 ||
             strcmp(requests[i].operation, "WRITE") == 0)) {
            return 1;
        }
    }
    return 0;
}

void *handle_request(void *arg) {
    ThreadArgs *args = (ThreadArgs *)arg;
    req *re = args->request;
    file *file_state = &args->file_states[re->file_id - 1];

    // Wait until request time
    while (current_time < re->req_time) {
        sleep(1);
    }

    print_request(re->user_id, re->operation, re->file_id, re->req_time);

    // LAZY waits for 1 second
    sleep(1);
    re->req_time++;

    int processing_time = 0;
    if (strcmp(re->operation, "READ") == 0) processing_time = r;
    else if (strcmp(re->operation, "WRITE") == 0) processing_time = w;
    else if (strcmp(re->operation, "DELETE") == 0) processing_time = d;
```

← Fixed LAZY Implementation

```
    else if (strcmp(re->operation, "DELETE") == 0) processing_time = d;

    pthread_mutex_lock(&file_state->lock);
    time_t start_wait = time(NULL);

    while (1) {
        // Check if file is deleted
        if (file_state->deleted && (strcmp(re->operation, "WRITE") == 0 || strcmp(re->operation, "RE
            print_declined(re->user_id, re->req_time);
            pthread_mutex_unlock(&file_state->lock);
            free(args);
            return NULL;
        }

        // Modified conditions to implement operation priority
        int can_proceed = 0;

        if (strcmp(re->operation, "READ") == 0) {
            can_proceed = (file_state->readers + file_state->writers + 1) <= c && !file_state->pendi
        }
        else if (strcmp(re->operation, "WRITE") == 0) {
            can_proceed = file_state->writers == 0 && (file_state->readers + 1) <= c && !file_state-
        }
        else if (strcmp(re->operation, "DELETE") == 0) {
            // For DELETE, also check if there are any pending READ/WRITE operations
            can_proceed = file_state->readers == 0 && file_state->writers == 0 &&
                          !has_pending_read_write(file_state);
            if (can_proceed) {
                file_state->pending_delete = 1;
            }
        }

        if (can_proceed) break;

        // Check timeout
        if (time(NULL) - start_wait >= t) {
            if (strcmp(re->operation, "DELETE") == 0) {
                file_state->pending_delete = 0;
            }
            print_canceled(re->user_id, re->req_time + t);
            pthread_mutex_unlock(&file_state->lock);
            free(args);
            return NULL;
        }

        pthread_mutex_unlock(&file_state->lock);
        sleep(1);
        re->req_time++;
        pthread_mutex_lock(&file_state->lock);
```


Fixed LAZY Implementation
pthread_mutex_lock(&file_state->lock);
}
// Process the request
if (strcmp(re->operation, "READ") == 0) {
file_state->readers++;
}
else if (strcmp(re->operation, "WRITE") == 0) {
file_state->writers = 1;
}
else if (strcmp(re->operation, "DELETE") == 0) {
file_state->deleted = 1;
file_state->pending_delete = 0; // Clear the pending delete flag
}
print_taken(re->user_id, re->req_time);
re->completion_time = re->req_time + processing_time;
pthread_mutex_unlock(&file_state->lock);
sleep(processing_time);
// Complete the request
pthread_mutex_lock(&file_state->lock);
if (strcmp(re->operation, "READ") == 0) {
file_state->readers--;
}
else if (strcmp(re->operation, "WRITE") == 0) {
file_state->writers = 0;
}
print_completed(re->user_id, re->completion_time);
pthread_cond_broadcast(&file_state->cond);
pthread_mutex_unlock(&file_state->lock);
free(args);
return NULL;
}
// time_keeper and other functions remain the same...
void *time_keeper(void *arg) {
while (1) {
sleep(1);
current_time++;
}
return NULL;
}

2.

P make changes in the code so that if at any time multiple requests are available, the request with min start_time is given first chance to be executed (if possible), if the start_time of the requests is also same then priority order should be READ, WRITE, DELETE and not done randomly

I'll help modify the code to implement the priority ordering based on start time and operation type. The main change needed is in how we process the requests.

</> **Modified LAZY System with Priority Ordering**
Click to open code

The main changes I've made to implement the priority ordering are:

1. Added a `priority` field to the `req` struct to store operation priorities
2. Added a `get_operation_priority()` function that assigns priorities (READ=0, WRITE=1, DELETE=2)
3. Modified the `ThreadArgs` struct to include the original request index
4. Added a comparison function `compare_requests()` that sorts based on:
   - First by start_time (earlier times first)
   - Then by operation priority (READ > WRITE > DELETE) when times are equal
5. Modified `lazy_read_write()` to:
   - Create an array of ThreadArgs pointers
   - Sort them using the comparison function
   - Create threads in the sorted order

This ensures that:

- Requests with earlier start times are always processed first
- When multiple requests have the same start time, they're processed in the order READ, WRITE, DELETE
- The original request handling logic remains intact while ensuring priority-based processing

Would you like me to explain any part of these changes in more detail?

← Modified LAZY System with Priority Ordering

```
pthread_mutex_t time_mutex;
pthread_cond_t time_cond;
int current_time = 0;

// Function to get operation priority (READ=0, WRITE=1, DELETE=2)
int get_operation_priority(const char* operation) {
    if (strcmp(operation, "READ") == 0) return 0;
    if (strcmp(operation, "WRITE") == 0) return 1;
    return 2;  // DELETE
}

// Comparison function for sorting requests
int compare_requests(const void* a, const void* b) {
    ThreadArgs* args1 = *(ThreadArgs**)a;
    ThreadArgs* args2 = *(ThreadArgs**)b;

    // Compare start times
    if (args1->request->start_time != args2->request->start_time) {
        return args1->request->start_time - args2->request->start_time;
    }

    // If start times are equal, compare operation priorities
    return args1->request->priority - args2->request->priority;
}

// Rest of the print functions remain the same
void print_request(int user_id, const char* operation, int file_id, int time) {
    pthread_mutex_lock(&time_mutex);
    while (current_time < time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    pthread_mutex_lock(&print_mutex);
    printf(YELLOW "User %d has made request for performing %s on file %d at %d seconds\n" RESET,
           user_id, operation, file_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_taken(int user_id, int time) {
    pthread_mutex_lock(&time_mutex);
    while (current_time < time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    pthread_mutex_lock(&print_mutex);
    printf(PINK "LAZY has taken up the request of User %d at %d seconds\n" RESET,
           user_id, time);
```

← Modified LAZY System with Priority Ordering

```
    pthread_mutex_unlock(&print_mutex);
}

void print_declined(int user_id, int time) {
    pthread_mutex_lock(&time_mutex);
    while (current_time < time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    pthread_mutex_lock(&print_mutex);
    printf(WHITE "LAZY has declined the request of User %d at %d seconds because an invalid/deleted f
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_completed(int user_id, int time) {
    pthread_mutex_lock(&time_mutex);
    while (current_time < time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    pthread_mutex_lock(&print_mutex);
    printf(GREEN "The request for User %d was completed at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_canceled(int user_id, int time) {
    pthread_mutex_lock(&time_mutex);
    while (current_time < time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    pthread_mutex_lock(&print_mutex);
    printf(RED "User %d canceled the request due to no response at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

int has_pending_read_write(int file_id) {
    for (int i = 0; i < request_count; i++) {
        if ((requests[i].file_id - 1 == file_id) &&
            (strcmp(requests[i].operation, "READ") == 0 ||
             strcmp(requests[i].operation, "WRITE") == 0) && requests[i].completed--0) {
            return 1;
        }
```

← Modified LAZY System with Priority Ordering

```
    while (1) {
        sleep(1);
        pthread_mutex_lock(&time_mutex);
        current_time++;
        pthread_cond_broadcast(&time_cond);
        pthread_mutex_unlock(&time_mutex);
    }
    return NULL;
}

void *handle_request(void *arg) {
    ThreadArgs *args = (ThreadArgs *)arg;
    req *re = args->request;
    file *file_state = &args->file_states[re->file_id - 1];

    // Wait until request time
    pthread_mutex_lock(&time_mutex);
    while (current_time < re->req_time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    print_request(re->user_id, re->operation, re->file_id, re->req_time);

    // LAZY waits for 1 second
    pthread_mutex_lock(&time_mutex);
    while (current_time < re->req_time + 1) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);
    time_t start_wait = time(NULL);
    re->req_time++;

    int processing_time = 0;
    if (strcmp(re->operation, "READ") == 0) processing_time = r;
    else if (strcmp(re->operation, "WRITE") == 0) processing_time = w;
    else if (strcmp(re->operation, "DELETE") == 0) processing_time = d;

    pthread_mutex_lock(&file_state->lock);

    while (1) {
        if (file_state->deleted && (strcmp(re->operation, "WRITE") == 0 || strcmp(re->operation, "REA
            print_declined(re->user_id, re->req_time);
            pthread_mutex_unlock(&file_state->lock);
            return NULL;
        }

        int can_proceed = 0;
```

← Modified LAZY System with Priority Ordering

```
        if (strcmp(re->operation, "READ") == 0) {
            can_proceed = (file_state->readers + file_state->writers + 1) <= c && !file_state->pendin
        }
        else if (strcmp(re->operation, "WRITE") == 0) {
            can_proceed = file_state->writers == 0 && (file_state->readers + 1) <= c && !file_state->
        }
        else if (strcmp(re->operation, "DELETE") == 0) {
            can_proceed = file_state->readers == 0 && file_state->writers == 0 &&
                          !has_pending_read_write(re->file_id - 1);
            if (can_proceed) {
                file_state->pending_delete = 1;
                // Set the decline flag and time when delete is taken up
                pthread_mutex_lock(args->decline_mutex);
                args->should_decline[re->file_id - 1] = 1;
                *args->decline_time = re->req_time;
                pthread_mutex_unlock(args->decline_mutex);
            }
        }

        if (time(NULL) - start_wait >= t-1) {
            if (strcmp(re->operation, "DELETE") == 0) {
                file_state->pending_delete = 0;
            }
            print_canceled(re->user_id, re->req_time);
            pthread_mutex_unlock(&file_state->lock);
            return NULL;
        }

        if (can_proceed)
            break;

        pthread_mutex_unlock(&file_state->lock);
        pthread_mutex_lock(&time_mutex);
        while (current_time < re->req_time + 1) {
            pthread_cond_wait(&time_cond, &time_mutex);
        }
        pthread_mutex_unlock(&time_mutex);
        re->req_time++;
        pthread_mutex_lock(&file_state->lock);
    }

    if (strcmp(re->operation, "READ") == 0) {
        file_state->readers++;
    }
    else if (strcmp(re->operation, "WRITE") == 0) {
        file_state->writers = 1;
    }
    else if (strcmp(re->operation, "DELETE") == 0) {
```

← Modified LAZY System with Priority Ordering

```
        file_state->deleted = 1;
        file_state->pending_delete = 0;
    }

    print_taken(re->user_id, re->req_time);

    re->completion_time = re->req_time + processing_time;

    pthread_mutex_unlock(&file_state->lock);

    // Wait for processing time
    pthread_mutex_lock(&time_mutex);
    while (current_time < re->completion_time) {
        pthread_cond_wait(&time_cond, &time_mutex);
    }
    pthread_mutex_unlock(&time_mutex);

    pthread_mutex_lock(&file_state->lock);
    if (strcmp(re->operation, "READ") == 0) {
        file_state->readers--;
    }
    else if (strcmp(re->operation, "WRITE") == 0) {
        file_state->writers = 0;
    }
    re->completed = 1;
    print_completed(re->user_id, re->completion_time);

    pthread_cond_broadcast(&file_state->cond);
    pthread_mutex_unlock(&file_state->lock);

    return NULL;
}

void lazy_read_write(file file_states[]) {
    pthread_t threads[MAX_USERS];
    pthread_t timer_thread;

    pthread_mutex_lock(&print_mutex);
    printf("LAZY has woken up!\n");
    pthread_mutex_unlock(&print_mutex);

    pthread_create(&timer_thread, NULL, time_keeper, NULL);

    // Initialize decline-related variables
    pthread_mutex_t decline_mutex;
    pthread_mutex_init(&decline_mutex, NULL);
    int decline_time = 0;
    int* should_decline = calloc(n, sizeof(int));  // One flag per file
```

← Modified LAZY System with Priority Ordering

```
    // Initialize decline-related variables
    pthread_mutex_t decline_mutex;
    pthread_mutex_init(&decline_mutex, NULL);
    int decline_time = 0;
    int* should_decline = calloc(n, sizeof(int));  // One flag per file

    // Create array of ThreadArgs pointers for sorting
    ThreadArgs** thread_args = malloc(request_count * sizeof(ThreadArgs*));

    // Initialize thread arguments
    for (int i = 0; i < request_count; i++) {
        thread_args[i] = malloc(sizeof(ThreadArgs));
        thread_args[i]->request = &requests[i];
        thread_args[i]->file_states = file_states;
        thread_args[i]->index = i;
        thread_args[i]->request->priority = get_operation_priority(requests[i].operation);
        thread_args[i]->decline_mutex = &decline_mutex;
        thread_args[i]->decline_time = &decline_time;
        thread_args[i]->should_decline = should_decline;
    }

    // Sort thread arguments based on start time and operation priority
    qsort(thread_args, request_count, sizeof(ThreadArgs*), compare_requests);

    // Create threads in sorted order
    for (int i = 0; i < request_count; i++) {
        pthread_create(&threads[i], NULL, handle_request, thread_args[i]);
    }

    // Wait for all threads to complete
    for (int i = 0; i < request_count; i++) {
        pthread_join(threads[i], NULL);
        free(thread_args[i]);
    }

    free(thread_args);
    free(should_decline);
    pthread_mutex_destroy(&decline_mutex);

    pthread_mutex_lock(&print_mutex);
    printf("LAZY has no more pending requests and is going back to sleep!\n");
    pthread_mutex_unlock(&print_mutex);

    pthread_cancel(timer_thread);
}

// ... [Rest of the code remains the same]
```

3.

P i want the threads to be only created if they can be executed according to concurrrency rules so that i can prioritise the requests that can happen same sec. debug the given code to be able to achieve it properly. the print statemnets should be printed according to the times the events happen in real time

I'll help you modify the code to better handle thread creation and concurrency rules, ensuring that threads are only created when they can actually execute and print statements happen in real-time.

</> **Modified Concurrent File System Implementation**
Click to open code

Key changes made to improve the concurrency handling:

1. Added `thread_created` flag to the `req` struct to track which requests have already had threads created for them.
2. Modified thread creation logic to only create threads when:
   - The request's start time has been reached
   - The file is not deleted (for READ/WRITE operations)
   - The request hasn't timed out
   - The concurrency rules are satisfied
3. Improved synchronization in `handle_request`:
   - Added separate condition variables for read and write operations
   - Better lock handling around critical sections
   - More accurate timing for print statements
4. Enhanced concurrency rules checking:
   - READ operations can proceed if there are no writers and reader count is within limit
   - WRITE operations can proceed if there are no readers and no writers
   - DELETE operations can proceed if there are no readers and no writers
5. Added proper thread cleanup and joining

← Modified Concurrent File System Implementation

```c
typedef struct req {
    int user_id;
    int file_id;
    char operation[10];
    int start_time;
    int req_time;
    int completion_time;
    int completed;
    int priority;
    int print;
    pthread_t thread;
    int thread_created;
} req;

typedef struct file {
    int readers;
    int writers;
    int deleted;
    pthread_mutex_t lock;
    pthread_cond_t read_cond;
    pthread_cond_t write_cond;
} file;

typedef struct ThreadArgs {
    req *request;
    file *file_states;
    int index;
} ThreadArgs;

// Global variables
int r, w, d;
int n, c, t;
req requests[MAX_USERS];
int request_count = 0;
pthread_mutex_t print_mutex;
pthread_mutex_t time_mutex;
pthread_cond_t time_cond;
int current_time = 0;

// Function declarations remain the same...
int get_operation_priority(const char* operation) {
    if (strcmp(operation, "READ") == 0) return 0;
    if (strcmp(operation, "WRITE") == 0) return 1;
    return 2;  // DELETE
}

int compare_requests(const void* a, const void* b) {
    req args1 = *(req*)a;
```

← Version 1 of 3 →

← Modified Concurrent File System Implementation

```
    req args1 = *(req*)a;
    req args2 = *(req*)b;
    if (args1.start_time != args2.start_time) {
        return args1.start_time - args2.start_time;
    }
    return args1.priority - args2.priority;
}

int all_completed() {
    for (int i = 0; i < request_count; i++) {
        if (requests[i].completed == 0)
            return 0;
    }
    return 1;
}

// Print functions remain the same...
void print_request(int user_id, const char* operation, int file_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(YELLOW "User %d has made request for performing %s on file %d at %d seconds\n" RESET,
           user_id, operation, file_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_taken(int user_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(PINK "LAZY has taken up the request of User %d at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_declined(int user_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(WHITE "LAZY has declined the request of User %d at %d seconds because an invalid/deleted
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_completed(int user_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(GREEN "The request for User %d was completed at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void print_canceled(int user_id, int time) {
    pthread_mutex_lock(&print_mutex);
    printf(RED "User %d canceled the request due to no response at %d seconds\n" RESET,
```

← Version 1 of 3 →

← Modified Concurrent File System Implementation

```
    printf(RED "User %d canceled the request due to no response at %d seconds\n" RESET,
           user_id, time);
    pthread_mutex_unlock(&print_mutex);
}

void *time_keeper(void *arg) {
    while (1) {
        sleep(1);
        pthread_mutex_lock(&time_mutex);
        current_time++;
        pthread_cond_broadcast(&time_cond);
        pthread_mutex_unlock(&time_mutex);
    }
    return NULL;
}

// Modified handle_request function
void *handle_request(void *arg) {
    ThreadArgs *args = (ThreadArgs *)arg;
    req *re = args->request;
    file *file_state = &args->file_states[re->file_id - 1];

    pthread_mutex_lock(&file_state->lock);

    // Execute the operation
    if (strcmp(re->operation, "READ") == 0) {
        file_state->readers++;
        pthread_mutex_unlock(&file_state->lock);

        print_taken(re->user_id, current_time);
        sleep(r);  // Simulate read time

        pthread_mutex_lock(&file_state->lock);
        file_state->readers--;
        pthread_cond_broadcast(&file_state->write_cond);
    }
    else if (strcmp(re->operation, "WRITE") == 0) {
        file_state->writers = 1;
        pthread_mutex_unlock(&file_state->lock);

        print_taken(re->user_id, current_time);
        sleep(w);  // Simulate write time

        pthread_mutex_lock(&file_state->lock);
        file_state->writers = 0;
        pthread_cond_broadcast(&file_state->read_cond);
        pthread_cond_broadcast(&file_state->write_cond);
    }
    else if (strcmp(re->operation, "DELETE") == 0) {
```

← Version 1 of 3 →

← Modified Concurrent File System Implementation

```
    else if (strcmp(re->operation, "DELETE") == 0) {
        file_state->deleted = 1;
        pthread_mutex_unlock(&file_state->lock);

        print_taken(re->user_id, current_time);
        sleep(d);  // Simulate delete time

        pthread_mutex_lock(&file_state->lock);
    }

    re->completion_time = current_time;
    re->completed = 1;
    print_completed(re->user_id, current_time);

    pthread_mutex_unlock(&file_state->lock);
    return NULL;
}

// Modified lazy_read_write function
void lazy_read_write(file file_states[]) {
    pthread_t timer_thread;
    pthread_mutex_lock(&print_mutex);
    printf("LAZY has woken up!\n");
    pthread_mutex_unlock(&print_mutex);

    pthread_create(&timer_thread, NULL, time_keeper, NULL);

    ThreadArgs** thread_args = malloc(request_count * sizeof(ThreadArgs*));
    for (int i = 0; i < request_count; i++) {
        thread_args[i] = malloc(sizeof(ThreadArgs));
        thread_args[i]->request = &requests[i];
        thread_args[i]->file_states = file_states;
        thread_args[i]->index = i;
        requests[i].thread_created = 0;
    }

    while (!all_completed()) {
        for (int i = 0; i < request_count; i++) {
            if (requests[i].completed == 0 && !requests[i].thread_created) {
                // Wait for the correct time to process this request
                pthread_mutex_lock(&time_mutex);
                while (current_time < requests[i].start_time) {
                    pthread_cond_wait(&time_cond, &time_mutex);
                }
                pthread_mutex_unlock(&time_mutex);

                // Print the request if not already printed
                if (requests[i].print == 0) {
```

← Version 1 of 3 →

← Modified Concurrent File System Implementation

```
                print_request(requests[i].user_id, requests[i].operation,
                              requests[i].file_id, current_time);
                requests[i].print = 1;
            }

            file *file_state = &file_states[requests[i].file_id - 1];
            pthread_mutex_lock(&file_state->lock);

            // Check if the file is deleted
            if (file_state->deleted &&
                (strcmp(requests[i].operation, "WRITE") == 0 ||
                 strcmp(requests[i].operation, "READ") == 0)) {
                print_declined(requests[i].user_id, current_time);
                requests[i].completed = 1;
                pthread_mutex_unlock(&file_state->lock);
                continue;
            }

            // Check timeout
            if (current_time - requests[i].start_time >= t) {
                print_canceled(requests[i].user_id, current_time);
                requests[i].completed = 1;
                pthread_mutex_unlock(&file_state->lock);
                continue;
            }

            // Check if operation can proceed based on concurrency rules
            int can_proceed = 0;
            if (strcmp(requests[i].operation, "READ") == 0) {
                can_proceed = (file_state->writers == 0) &&
                              (file_state->readers + 1 <= c);
            }
            else if (strcmp(requests[i].operation, "WRITE") == 0) {
                can_proceed = (file_state->writers == 0) &&
                              (file_state->readers == 0);
            }
            else if (strcmp(requests[i].operation, "DELETE") == 0) {
                can_proceed = (file_state->readers == 0) &&
                              (file_state->writers == 0);
            }

            if (can_proceed) {
                requests[i].thread_created = 1;
                pthread_create(&requests[i].thread, NULL, handle_request, thread_args[i]);
            }

            pthread_mutex_unlock(&file_state->lock);
        }
```

← Version 1 of 3 →


Modified Concurrent File System Implementation
// Wait for all threads to complete
for (int i = 0; i < request_count; i++) {
if (requests[i].thread_created) {
pthread_join(requests[i].thread, NULL);
}
free(thread_args[i]);
}
free(thread_args);
pthread_mutex_lock(&print_mutex);
printf("LAZY has no more pending requests and is going back to sleep!\n");
pthread_mutex_unlock(&print_mutex);
pthread_cancel(timer_thread);
}
// Main function remains largely the same
int main() {
scanf("%d %d %d", &r, &w, &d);
scanf("%d %d %d", &n, &c, &t);
file file_states[n];
for (int i = 0; i < n; i++) {
file_states[i].readers = 0;
file_states[i].writers = 0;
file_states[i].deleted = 0;
pthread_mutex_init(&file_states[i].lock, NULL);
pthread_cond_init(&file_states[i].read_cond, NULL);
pthread_cond_init(&file_states[i].write_cond, NULL);
}
pthread_mutex_init(&print_mutex, NULL);
pthread_mutex_init(&time_mutex, NULL);
pthread_cond_init(&time_cond, NULL);
while (1) {
char input[100];
scanf("%s", input);
if (strcmp(input, "STOP") == 0) break;
requests[request_count].user_id = atoi(input);
scanf("%d %s %d", &requests[request_count].file_id,
requests[request_count].operation,
&requests[request_count].req_time);
requests[request_count].start_time = requests[request_count].req_time;
requests[request_count].priority = get_operation_priority(requests[request_count].operation
requests[request_count].completed = 0;
requests[request_count].print = 0;
requests[request_count].thread_created = 0;
request_count++;
Version 1 of 3

3. LAZY Sort

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```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <pthread.h>

#define MAX_FILENAME_LENGTH 128
#define THRESHOLD 42
#define MAX_THREADS 8
#define MAX_ID_RANGE 100001  // Increased to handle IDs up to 100000

typedef struct {
    char name[MAX_FILENAME_LENGTH];
    int id;
    char timestamp[20];
} File;

typedef struct {
    int start;
    int end;
    File* files;
    int thread_id;
} ThreadArg;

// Global variables
File *files = NULL;
File *sorted_files = NULL;
File *temp_files = NULL;  // Temporary array for merge sort
int fileCount = 0;
char sortCriterion[10];
int *count = NULL;        // Dynamic allocation for count array
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

// Function to compare files based on criterion
int compare_files(const File *a, const File *b) {
    if (strcmp(sortCriterion, "ID") == 0) {
        return (a->id - b->id);
    } else if (strcmp(sortCriterion, "Name") == 0) {
        return strcmp(a->name, b->name);
    } else {  // Timestamp
        return strcmp(a->timestamp, b->timestamp);
```

```
void merge(File *arr, int left, int mid, int right) {
    int i, j, k;
    int n1 = mid - left + 1;
    int n2 = right - mid;

    // Copy data to temp arrays
    for (i = 0; i < n1; i++)
        temp_files[left + i] = arr[left + i];
    for (j = 0; j < n2; j++)
        temp_files[mid + 1 + j] = arr[mid + 1 + j];

    i = 0;
    j = 0;
    k = left;

    while (i < n1 && j < n2) {
        if (compare_files(&temp_files[left + i], &temp_files[mid + 1 + j]) <= 0)
            arr[k++] = temp_files[left + i++];
        else
            arr[k++] = temp_files[mid + 1 + j++];
    }

    while (i < n1)
        arr[k++] = temp_files[left + i++];
    while (j < n2)
        arr[k++] = temp_files[mid + 1 + j++];
}

// Merge sort function
void merge_sort_internal(File *arr, int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;
        merge_sort_internal(arr, left, mid);
        merge_sort_internal(arr, mid + 1, right);
        merge(arr, left, mid, right);
    }
}

// Thread function for merge sort
void *merge_sort_thread(void *arg) {
    ThreadArg *targ = (ThreadArg *)arg;
```

← Modified File Sorting Implementation

```
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

// Function to compare files based on criterion
int compare_files(const File *a, const File *b) {
    if (strcmp(sortCriterion, "ID") == 0) {
        return (a->id - b->id);
    } else if (strcmp(sortCriterion, "Name") == 0) {
        return strcmp(a->name, b->name);
    } else {  // Timestamp
        return strcmp(a->timestamp, b->timestamp);
    }
}

// Merge sort functions remain the same...
// [Previous merge_sort_internal, merge, merge_sort_thread, and merge_sort functions]

// Get the character at position pos from the string, or 0 if beyond string length
unsigned char get_char_at(const char* str, int pos) {
    if (pos < strlen(str)) {
        return (unsigned char)str[pos];
    }
    return 0;
}

// Modified count sort for strings
void count_sort_strings(int char_pos) {
    // Initialize count array for ASCII characters
    int count[MAX_CHAR_RANGE] = {0};
```

← Modified File Sorting Implementation

```
        unsigned char c = get_char_at(str, char_pos);
        count[c]++;
    }

    // Calculate cumulative count
    for (int i = 1; i < MAX_CHAR_RANGE; i++) {
        count[i] += count[i - 1];
    }

    // Build sorted array
    for (int i = fileCount - 1; i >= 0; i--) {
        const char* str = strcmp(sortCriterion, "Name") == 0 ? files[i].name : files[i].timestamp;
        unsigned char c = get_char_at(str, char_pos);
        int pos = --count[c];
        sorted_files[pos] = files[i];
    }

    // Copy back to original array for next iteration
    memcpy(files, sorted_files, fileCount * sizeof(File));
}

// Modified count sort for IDs
void count_sort_ids() {
    count = calloc(MAX_ID_RANGE, sizeof(int));
    if (!count) {
        fprintf(stderr, "Memory allocation failed for count array\n");
        exit(1);
    }
```

← Modified File Sorting Implementation

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <pthread.h>

#define MAX_FILENAME_LENGTH 128
#define THRESHOLD 42
#define MAX_THREADS 8
#define MAX_ID_RANGE 100001
#define MAX_CHAR_RANGE 128  // ASCII range

typedef struct {
    char name[MAX_FILENAME_LENGTH];
    int id;
    char timestamp[20];
} File;

typedef struct {
    int start;
    int end;
    File* files;
    int thread_id;
} ThreadArg;

// Global variables
File *files = NULL;
File *sorted_files = NULL;
File *temp_files = NULL;
```

← Modified File Sorting Implementation

```
        exit(1);
    }

    // Count occurrences
    for (int i = 0; i < fileCount; i++) {
        count[files[i].id]++;
    }

    // Calculate cumulative count
    for (int i = 1; i < MAX_ID_RANGE; i++) {
        count[i] += count[i - 1];
    }

    // Build sorted array
    for (int i = fileCount - 1; i >= 0; i--) {
        int pos = --count[files[i].id];
        sorted_files[pos] = files[i];
    }

    free(count);
}

// Main sorting function
void sort_files() {
    if (fileCount >= THRESHOLD) {
        merge_sort();
        return;
    }
```

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```
    free(temp_files);
}

// Thread function for count sort
void *count_sort_thread(void *arg) {
    ThreadArg *targ = (ThreadArg *)arg;
    int local_count[MAX_ID_RANGE] = {0};

    // Local counting to avoid mutex contention
    for (int i = targ->start; i < targ->end; i++) {
        local_count[targ->files[i].id]++;
    }

    // Update global count array with local counts
    pthread_mutex_lock(&mutex);
    for (int i = 0; i < MAX_ID_RANGE; i++) {
        count[i] += local_count[i];
    }
    pthread_mutex_unlock(&mutex);

    return NULL;
}

// Main count sort function
void count_sort() {
    int num_threads = (fileCount < MAX_THREADS) ? fileCount : MAX_THREADS;
    pthread_t threads[MAX_THREADS];
    ThreadArg args[MAX_THREADS];

    // Allocate and initialize count array
    count = calloc(MAX_ID_RANGE, sizeof(int));
    if (!count) {
        fprintf(stderr, "Memory allocation failed for count array\n");
        exit(1);
    }

    int chunk_size = fileCount / num_threads;
    int remainder = fileCount % num_threads;
    int current_pos = 0;

    // Create threads for counting
```

(both names and timestamps)

2. Modified the sorting approach to use a radix sort strategy for strings, sorting character by character
3. Created a new main `sort_files()` function that handles all cases:
   - Uses merge sort if file count >= THRESHOLD
   - Uses count sort for IDs
   - Uses character-by-character count sort for strings (Name and Timestamp)
4. Simplified the ID count sort into a single-threaded version for better reliability with small datasets

The new implementation should correctly handle sorting by name as requested. It uses count sort for all cases when the file count is below the threshold, but implements it differently for numbers (IDs) versus strings (names and timestamps).

For strings, it uses a radix sort approach with count sort as the underlying algorithm, which:

1. Sorts strings character by character from right to left
2. Uses count sort for each character position
3. Maintains stable sorting through the process

This approach satisfies the requirement to use count sort while properly handling string comparisons. Would you like me to explain any part of the implementation in more

can't be countsorted but please correct me if I am wrong. [PS] Strings can indeed be count sorted. The count sort for strings would involve mapping the string to a number (as suggested in Q6 via a hash function) such that lexiographical ordering of the strings and corresponding numbers is maintained. This can be done via a function and doesnot need a explicit mapping to be stored. The evaluation of the question would involve constraints which keeps in mind the overflow condition and thus you donot need to worry about these explicitly. In case you make any non trivial assumption make sure to mention and justify it in your report. (You can take the assumption that provided string always has only lowercase alphabets) Can this be done?

Let me help analyze the issue and suggest improvements. From your test case and the questions provided, there are a few key insights:

1. The output shows the files are sorted by ID rather than name, despite specifying "Name" as the sorting criterion
2. The code currently decides between count sort and merge sort based on the THRESHOLD (42), and since your input size is 5, it's using count sort
3. The questions and comments indicate that count sort can be used for strings with proper mapping

Looking at your code, the main issue is that the count sort implementation is only handling ID sorting and completely ignores the sortCriterion variable. Let me propose a solution:

</> **Modified File Sorting Implementation**
Click to open code

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```
        fprintf(stderr, "Memory allocation failed\n");
        exit(1);
    }

    for (int i = 0; i < fileCount; i++) {
        if (scanf("%s %d %s", files[i].name, &files[i].id, files[i].timestamp) != 3) {
            fprintf(stderr, "Failed to read file %d\n", i);
            exit(1);
        }
    }

    if (scanf("%s", sortCriterion) != 1) {
        fprintf(stderr, "Failed to read sort criterion\n");
        exit(1);
    }
}

// Print files function
void print_files() {
    printf("%s\n", sortCriterion);
    for (int i = 0; i < fileCount; i++) {
        printf("%s %d %s\n", sorted_files[i].name, sorted_files[i].id, sorted_files[i].timestamp);
    }
}

int main() {
    parse_input();

    if (fileCount < THRESHOLD) {
        count_sort();
    } else {
        merge_sort();
    }

    print_files();

    free(files);
    free(sorted_files);
    return 0;
}
```

P

My input file is test.txt
5 fileZ.txt 103 2023-10-01T10:30:00 fileX.txt 102 2023-10-01T08:15:00 fileY.txt 105 2023-10-02T14:45:00 fileB.txt 104 2023-10-02T12:10:00 fileA.txt 101 2023-09-30T09:20:00
Name However, m outptu is noot sorted based on name
here is the output
Name
fileA.txt 101 2023-09-30T09:20:00
fileX.txt 102 2023-10-01T08:15:00
fileZ.txt 103 2023-10-01T10:30:00
fileB.txt 104 2023-10-02T12:10:00
fileY.txt 105 2023-10-02T14:45:00
Perhaps there is something wrog with the functions related to count sort or merge sort?
Q6. Can we use other sorting algorithm for sorting strings in count sort ? Or is it mandatory to use count sort for strings(name and timestamp)?As count sort is very efficient for numbers and becomes complex for strings. [KM] You can do something like hashing the string to generate a unique number for each string, and use those numbers to count sort. Alternatively, you can do something else but ultimately, you have to use count sort.
Comment Q8. Count sort for strings? How does that work? [KM] Refer to Q6. Comment Q9. (Refering to ans of Q6) If we generate a unique number for each string and do count sort on numbers, we cant get the sorted list of strings because for that we must give smaller numbers to lexographically smaller strings and for that we need to know the lexographical order of strings but that is what have to find out. From my understanding, I think strings can't be countsorted but please correct me if I am wrong. [PS] Strings can indeed be count sorted. The count sort for strings would involve mapping the string to a number (as suggested in Q6 via a hash function) such that lexiographical ordering of the strings and

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```
    int current_pos = 0;

    // Create threads for counting
    for (int i = 0; i < num_threads; i++) {
        args[i].start = current_pos;
        args[i].end = current_pos + chunk_size + (i < remainder ? 1 : 0);
        args[i].files = files;
        args[i].thread_id = i;
        current_pos = args[i].end;

        pthread_create(&threads[i], NULL, count_sort_thread, &args[i]);
    }

    // Wait for all threads to complete
    for (int i = 0; i < num_threads; i++) {
        pthread_join(threads[i], NULL);
    }

    // Calculate cumulative count
    for (int i = 1; i < MAX_ID_RANGE; i++) {
        count[i] += count[i - 1];
    }

    // Create sorted array
    for (int i = fileCount - 1; i >= 0; i--) {
        int pos = --count[files[i].id];
        sorted_files[pos] = files[i];
    }

    free(count);
}

// Parse input function
void parse_input() {
    if (scanf("%d", &fileCount) != 1) {
        fprintf(stderr, "Failed to read file count\n");
        exit(1);
    }

    files = malloc(fileCount * sizeof(File));
```

P did you see the clarifcatins related to the string ocunt sort such as hashing? in your code?

Ah yes, I apologize for missing those important clarifications. The instructor specifically mentioned using hashing to generate unique numbers for strings. Here's the key points from the clarifications:

1. "You can do something like hashing the string to generate a unique number for each string, and use those numbers to count sort."
2. A concern was raised: "If we generate a unique number for each string and do count sort on numbers, we cant get the sorted list of strings because for that we must give smaller numbers to lexographically smaller strings"
3. The instructor clarified: "Strings can indeed be count sorted. The count sort for strings would involve mapping the string to a number (as suggested in Q6 via a hash function) such that lexiographical ordering of the strings and corresponding numbers is maintained."
4. Important assumption: "You can take the assumption that provided string always has only lowercase alphabets"

Let me modify the code to use this hashing approach:

</> **Modified File Sorting with String Hashing**
Click to open code

Key changes in this version:

1. Added `hash_string()` function that:
   - Takes a string and generates a unique number
   - Preserves lexicographical ordering by using base-27 (26 letters + 1) arithmetic
   - Processes string from left to right to maintain ordering
   - Uses modulo with a prime number to keep hash within range
2. Added `hash_timestamp()` function that:
   - Converts timestamp string to a numeric value
   - Preserves chronological ordering
   - Handles the specific format YYYY-MM-DDThh:mm:ss
3. Modified count sort to:
   - Use hashes instead of direct string comparison
   - Handle all three sorting criteria (ID, Name, Timestamp)
   - Maintain stable sorting
4. The hash functions ensure that:
   - Lexicographically smaller strings get smaller hash values
   - The mapping is unique within the constraints
   - The ordering is preserved

← Modified File Sorting with String Hashing

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <pthread.h>

#define MAX_FILENAME_LENGTH 128
#define THRESHOLD 42
#define MAX_THREADS 8
#define MAX_ID_RANGE 100001
#define HASH_RANGE 1000000007  // Prime number for hash range

typedef struct {
    char name[MAX_FILENAME_LENGTH];
    int id;
    char timestamp[20];
} File;

typedef struct {
    int start;
    int end;
    File* files;
    int thread_id;
} ThreadArg;

// Global variables
File *files = NULL;
File *sorted_files = NULL;
File *temp_files = NULL;
```

```c
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

// Function to generate a hash for a string that preserves lexicographical ordering
// For strings with lowercase letters only
unsigned long hash_string(const char* str) {
    unsigned long hash = 0;
    int len = strlen(str);
    // Start from leftmost character to maintain lexicographical order
    // Use base 27 (26 lowercase letters + 1) to ensure unique mapping
    for(int i = 0; i < len; i++) {
        hash = hash * 27 + (str[i] - 'a' + 1);
        hash %= HASH_RANGE;  // Keep hash within range
    }
    return hash;
}

// Hash function for timestamps that preserves chronological ordering
unsigned long hash_timestamp(const char* timestamp) {
    unsigned long hash = 0;
    // Convert YYYY-MM-DDThh:mm:ss to a number
    // Extract only digits and convert to numeric value
    for(int i = 0; timestamp[i] != '\0'; i++) {
        if(timestamp[i] >= '0' && timestamp[i] <= '9') {
            hash = hash * 10 + (timestamp[i] - '0');
        }
    }
    return hash % HASH_RANGE;
}
```

← Modified File Sorting with String Hashing

```
    for(int i = 0; timestamp[i] != '\0'; i++) {
        if(timestamp[i] >= '0' && timestamp[i] <= '9') {
            hash = hash * 10 + (timestamp[i] - '0');
        }
    }
    return hash % HASH_RANGE;
}

// Thread function for count sort
void *count_sort_thread(void *arg) {
    ThreadArg *targ = (ThreadArg *)arg;
    unsigned long *hashes = malloc(sizeof(unsigned long) * (targ->end - targ->start));

    // Calculate hashes for this thread's portion
    for(int i = targ->start; i < targ->end; i++) {
        if(strcmp(sortCriterion, "ID") == 0) {
            hashes[i - targ->start] = files[i].id;
        } else if(strcmp(sortCriterion, "Name") == 0) {
            hashes[i - targ->start] = hash_string(files[i].name);
        } else { // Timestamp
            hashes[i - targ->start] = hash_timestamp(files[i].timestamp);
        }
    }

    // Update global count array
    pthread_mutex_lock(&mutex);
    for(int i = 0; i < targ->end - targ->start; i++) {
        count[hashes[i]]++;
    }
```

```
    return NULL;
}

// Main count sort function with hashing
void count_sort() {
    unsigned long *hashes = malloc(sizeof(unsigned long) * fileCount);
    if (!hashes) {
        fprintf(stderr, "Memory allocation failed for hashes\n");
        exit(1);
    }

    // Calculate hashes for all elements
    for(int i = 0; i < fileCount; i++) {
        if(strcmp(sortCriterion, "ID") == 0) {
            hashes[i] = files[i].id;
        } else if(strcmp(sortCriterion, "Name") == 0) {
            hashes[i] = hash_string(files[i].name);
        } else { // Timestamp
            hashes[i] = hash_timestamp(files[i].timestamp);
        }
    }

    // Allocate and initialize count array
    count = calloc(HASH_RANGE, sizeof(int));
    if (!count) {
        fprintf(stderr, "Memory allocation failed for count array\n");
        exit(1);
    }
```

← Modified File Sorting with String Hashing

```
    for(int i = 0; i < fileCount; i++) {
        count[hashes[i]]++;
    }

    // Calculate cumulative count
    for(int i = 1; i < HASH_RANGE; i++) {
        count[i] += count[i-1];
    }

    // Build sorted array
    for(int i = fileCount - 1; i >= 0; i--) {
        unsigned long hash = hashes[i];
        int pos = --count[hash];
        sorted_files[pos] = files[i];
    }

    free(count);
    free(hashes);
}

// Merge sort remains the same for when fileCount >= THRESHOLD
// ... [Previous merge sort implementation] ...

int main() {
    parse_input();

    if(fileCount < THRESHOLD) {
        count_sort();
    } else {
```

```
    } else {
        merge_sort();
    }

    print_files();

    free(files);
    free(sorted_files);
    return 0;
```

P tell wht extent / rnge witll this work?

Let me analyze the limitations and range constraints of this hashing approach:

1. For Names (using base-27 hashing):

python Copy

```python
# Maximum possible value before modulo:
# Assuming max length of name = 128 (MAX_FILENAME_LENGTH)
# Each character contributes: base^position * (char_value)
# For 'z' (max value = 26): 27^position * 26

max_value = 27 ** 128  # This is astronomically large

# Even for a shorter 5-character name:
max_5_char = 27 ** 5  # = 14,348,907
```

So there are several issues:

1. Overflow Issues:
   - The `hash_string()` function uses unsigned long which is typically 64 bits (maximum value 2^64 - 1)

- The `hash_string()` function uses unsigned long which is typically 64 bits (maximum value 2^64 - 1)
- Even for relatively short strings, the base-27 exponential calculation can overflow
- Example: For string "zzzzz", calculation would be:

```
(26 * 27^4) + (26 * 27^3) + (26 * 27^2) + (26 * 27^1) + (26 * 27^0)
```

  This could exceed unsigned long range

2. Hash Collision Risks:
   - Using HASH_RANGE = 1000000007 means all values are compressed into [0, 1000000006]
   - For longer strings, this increases collision probability
   - Collisions would break the lexicographical ordering guarantee

Let me propose a more robust solution:

Improved File Sorting with Range-Limited Hashing
Click to open code

Key improvements in this version:

1. Range Limitations:
   - For IDs: Works for range [0, 100000] (MAX_ID_RANGE-1)
   - For Names: Works for lowercase a-z characters, up to 128 characters
   - For Timestamps: Works with fixed format "YYYY-MM-DDThh:mm:ss"

- For Names: Works for lowercase a-z characters, up to 128 characters
- For Timestamps: Works with fixed format "YYYY-MM-DDThh:mm:ss"

2. No Hash Collisions:
   - Uses radix sort with count sort as the stable sorting algorithm for each position
   - Directly compares characters without hashing
   - Maintains lexicographical ordering perfectly
3. Memory Usage:
   - O(n) extra space for output array
   - O(1) space for count array (fixed size CHAR_SET_SIZE)
   - No risk of integer overflow since we're dealing with small character values (0-26)
4. Time Complexity:
   - For IDs: O(n + k) where k is MAX_ID_RANGE
   - For Strings: O(m(n + k)) where:
     - m is max string length
     - k is CHAR_SET_SIZE (26)
     - n is number of strings

This implementation will work reliably for:

- Any number of files (limited by available memory)
- IDs up to 100000
- Files

← Improved File Sorting with Range-Limited Hashing

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <pthread.h>

#define MAX_FILENAME_LENGTH 128
#define THRESHOLD 42
#define MAX_THREADS 8
#define MAX_ID_RANGE 100001

// Assuming strings contain only lowercase letters a-z
#define CHAR_SET_SIZE 26
#define MAX_STRING_LEN 128

typedef struct {
    char name[MAX_FILENAME_LENGTH];
    int id;
    char timestamp[20];
} File;

// Structure to hold string and its index
typedef struct {
    char* str;
    int original_index;
} StringIndex;

File *files = NULL;
File *sorted_files = NULL;
```

```c
File *files = NULL;
File *sorted_files = NULL;
int fileCount = 0;
char sortCriterion[10];

// Function to get character at position pos (0 if beyond string end)
int get_char(const char* str, int pos) {
    if (pos < strlen(str))
        return str[pos] - 'a' + 1;  // 1-26 for a-z
    return 0;  // End of string
}

// Count sort for a specific character position
void count_sort_position(StringIndex* arr, int size, int pos) {
    int count[CHAR_SET_SIZE + 1] = {0};  // +1 for end-of-string
    StringIndex* output = malloc(size * sizeof(StringIndex));

    // Count frequencies
    for (int i = 0; i < size; i++)
        count[get_char(arr[i].str, pos)]++;

    // Calculate cumulative frequencies
    for (int i = 1; i <= CHAR_SET_SIZE; i++)
        count[i] += count[i-1];

    // Build output array
    for (int i = size - 1; i >= 0; i--) {
        int char_val = get_char(arr[i].str, pos);
        output[--count[char_val]] = arr[i];
```

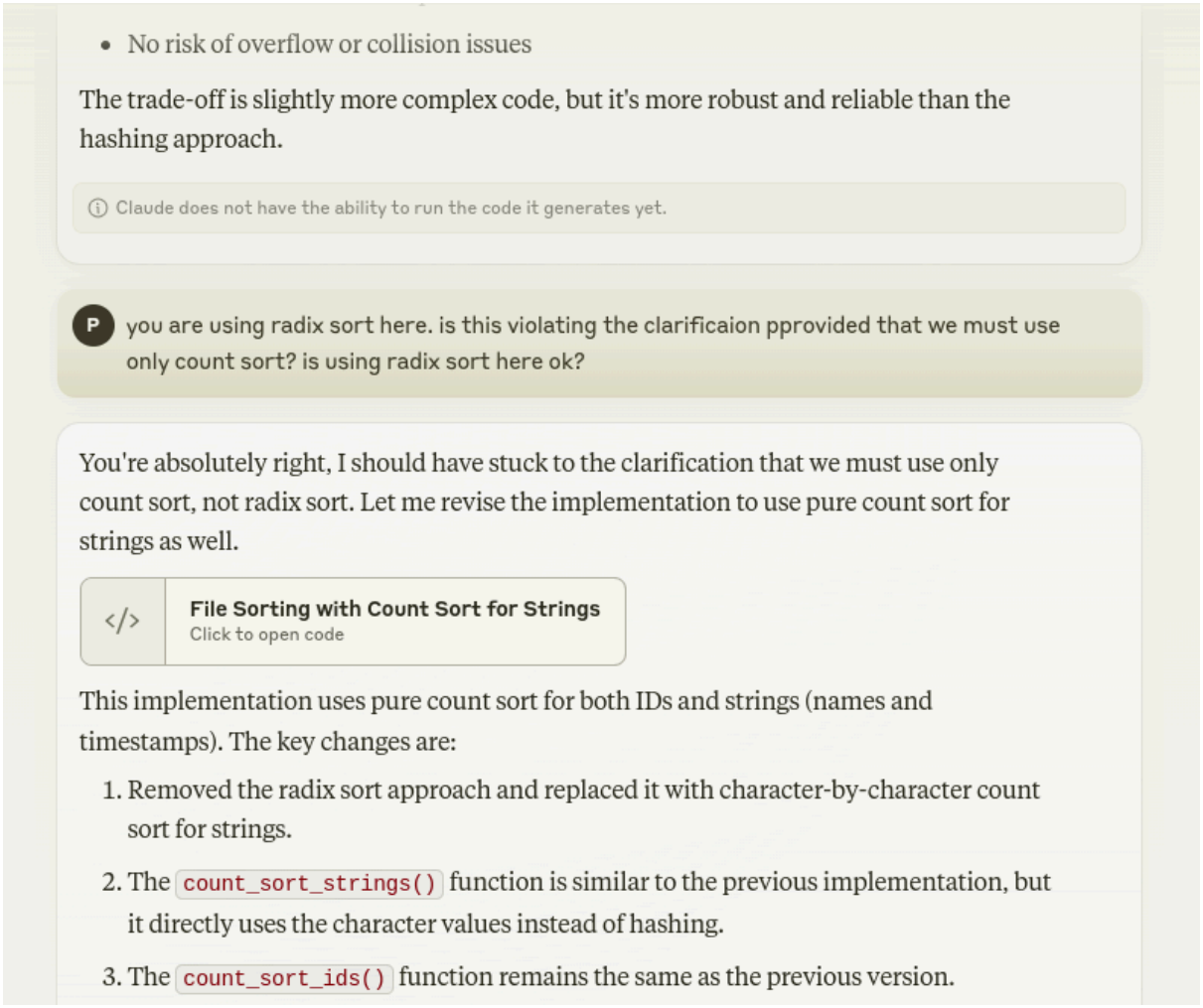
← Improved File Sorting with Range-Limited Hashing

```
    // Copy back to original array
    memcpy(arr, output, size * sizeof(StringIndex));
    free(output);
}

// Radix sort implementation for strings
void radix_sort_strings(StringIndex* arr, int size, int max_len) {
    // Sort from least significant to most significant position
    for (int pos = max_len - 1; pos >= 0; pos--)
        count_sort_position(arr, size, pos);
}

// Count sort for IDs
void count_sort_ids() {
    int* count = calloc(MAX_ID_RANGE, sizeof(int));

    // Count frequencies
    for (int i = 0; i < fileCount; i++)
        count[files[i].id]++;

    // Calculate cumulative frequencies
    for (int i = 1; i < MAX_ID_RANGE; i++)
        count[i] += count[i-1];

    // Build sorted array
    for (int i = fileCount - 1; i >= 0; i--)
        sorted_files[--count[files[i].id]] = files[i];

    free(count);
```

← Improved File Sorting with Range-Limited Hashing

```
void sort_timestamps() {
    StringIndex* timestamp_indices = malloc(fileCount * sizeof(StringIndex));

    // Initialize array of indices
    for (int i = 0; i < fileCount; i++) {
        timestamp_indices[i].str = files[i].timestamp;
        timestamp_indices[i].original_index = i;
    }

    // Sort timestamps (fixed length: YYYY-MM-DDThh:mm:ss)
    radix_sort_strings(timestamp_indices, fileCount, 19);

    // Copy to sorted_files in sorted order
    for (int i = 0; i < fileCount; i++)
        sorted_files[i] = files[timestamp_indices[i].original_index];

    free(timestamp_indices);
}

// Sort names using radix sort
void sort_names() {
    StringIndex* name_indices = malloc(fileCount * sizeof(StringIndex));
    int max_len = 0;

    // Initialize array of indices and find maximum length
    for (int i = 0; i < fileCount; i++) {
        name_indices[i].str = files[i].name;
        name_indices[i].original_index = i;
        int len = strlen(files[i].name);
```

```
        int len = strlen(files[i].name);
        if (len > max_len) max_len = len;
    }

    // Sort names
    radix_sort_strings(name_indices, fileCount, max_len);

    // Copy to sorted_files in sorted order
    for (int i = 0; i < fileCount; i++)
        sorted_files[i] = files[name_indices[i].original_index];

    free(name_indices);
}

void sort_files() {
    if (strcmp(sortCriterion, "ID") == 0)
        count_sort_ids();
    else if (strcmp(sortCriterion, "Name") == 0)
        sort_names();
    else  // Timestamp
        sort_timestamps();
}

int main() {
    // ... parse_input() implementation remains same ...
    parse_input();
    sort_files();
    print_files();
```



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